The primary text for Algebra II is Dummit and Foote's *Abstract Algebra*, and will cover the following topics:

- modules and advanced linear algebra;
- representation theory of finite groups;
- field theory and Galois theory.

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Modules and Advanced Linear Algebra

Tensor Products of Modules

The first major topic in Modules and Advanced Linear Algebra is tensor products.

Introduction and Basic Definitions

To motivate tensor products, we recall a basic fact from linear algebra. If we assume that R is a field, and M, N are finite-dimensional R-vector spaces, then the following equation necessarily holds:

$$\dim(M \oplus N) = \dim(M) + \dim(N).$$

We want to construct a similar operation on vector spaces, $M \otimes N$, that satisfies

$$\dim(M \otimes N) = \dim(M)\dim(N).$$

For now, we will label this by $M\bar{\otimes}N$, where we use the $\bar{\otimes}$ to refer to the fact that this is a temporary definition. Naively, we might seek to define $M\bar{\otimes}N$ as follows. If we let $\{x_1,\dots,x_k\}$ be a basis for M and $\{y_1,\dots,y_\ell\}$ a basis for N, then we will define $M\bar{\otimes}N$ to be all the formal R-linear combinations over the basis

$$B=\left\{x_i\otimes y_j \;\middle|\; 1\leq i\leq k, 1\leq j\leq \ell\right\}.$$

While this is technically correct — as in, this does yield a vector space with

$$\dim(M\bar{\otimes}N)=\dim(M)\dim(N),$$

the issue is that this definition is not canonical, in that it depends on chosen bases for M and N. Furthermore, it is not clear how one may generalize from this definition to modules over arbitrary rings, which do not necessarily have bases. To resolve this issue, we will go about defining a construction that "extends," in a sense, this definition of $M\bar{\otimes}N$.

To start, we define the simple tensor $m\otimes n$ for any $m\in M$ and $n\in N$. If we let

$$\begin{aligned} m&=\sum_{i=1}^{k}\lambda_i x_i\\ n&=\sum_{j=1}^{\ell}\mu_j y_j, \end{aligned}$$

then we will define

$$m\otimes n=\sum_{i=1}^{k}\sum_{j=1}^{\ell}\lambda_i\mu_j\left(x_i\otimes y_j\right).$$

We observe that every element of $M\bar{\otimes}N$ is a sum (i.e., an *integral* linear combination) of simple tensors, as by regrouping we may take

$$\sum_{i=1}^{k}\sum_{j=1}^{\ell}\lambda_{ij}\left(x_i\otimes y_j\right)=\sum_{i=1}^{k}\sum_{j=1}^{\ell}\left(\lambda_{ij}x_i\right)\otimes y_j.$$

The simple tensors satisfy the following relations:

(R1) $(m_1+m_2)\otimes n=m_1\otimes n+m_2\otimes n$;

(R2) $m\otimes(n_1+n_2)=m\otimes n_1+m\otimes n_2$;

(R3) $(\alpha m)\otimes n=m\otimes(\alpha n)$

for $m,m_1,m_2\in M$, $n,n_1,n_2\in N$, and $\alpha\in R$.

Proposition: These are the defining relations for $M\bar{\otimes}N$ in the category of abelian groups.

We will simply take this proposition as fact.

Now, let

$$\begin{aligned} Q&=M\times N\\ &=\{(m,n)\mid m\in M, n\in N\} \end{aligned}$$

be the Cartesian product of M and N as sets. We will then take $\mathbb{Z}[Q]$ to be the standard free $\mathbb{Z}$-module (i.e., free abelian group) on Q. That is, $\mathbb{Z}[Q]$ is the set of formal linear combinations

$$v=\sum_{q\in Q}\lambda_q q,$$

where $\lambda_q \in \mathbb{Z}$ and only finitely many coefficients are nonzero. By the universal property of free abelian groups, the map $(m,n) \mapsto m \otimes n$ descends to a unique homomorphism $\varphi\colon \mathbb{Z}[Q] \to M\bar{\otimes}N$. Such a homomorphism is necessarily surjective as every element of $M\bar{\otimes}N$ is an integral linear combination of simple tensors, meaning that we have

$$M\bar{\otimes}N \cong \mathbb{Z}[Q]/\ker(\varphi)$$

as abelian groups.

Now, consider the subgroup of $\mathbb{Z}[Q]$, which we denote $\langle K\rangle$, that is generated by the following elements:

(I) $(m_1+m_2,n)-(m_1,n)-(m_2,n)$;

(II) $(m,n_1+n_2)-(m,n_1)-(m,n_2)$;

(III) $(\alpha m,n)-(m,\alpha n)$

for $m_1,m_2,m \in M$, $n_1,n_2,n \in N$, and $\alpha \in R$. Then, from proposition that the relations (R1) through (R3) define $M\bar{\otimes}N$, it follows that $\langle K\rangle = \ker(\varphi)$. Thus, we may define the tensor product canonically as follows.

Definition: Letting M,N,Q,K be as above, we define

$$M \otimes N := \mathbb{Z}[Q]/\langle K\rangle, \tag{$\dagger$}$$

and define $m \otimes n = (m,n)+K$.

So far, this has only given us an abelian group. We may ask how to define $\mathbb{Z}[Q]/\langle K\rangle$ as an R-vector space, which naturally seems to be defined by

$$r\left(\sum_{i=1}^{n} m_i \otimes n_i\right) = \sum_{i=1}^{n} (rm_i)\otimes n_i \tag{$*$}$$

To show that the right-hand side of $(*)$ is well-defined is a very difficult task. We will not do it here.

Now, we can actually quite easily generalize $(\dagger)$ to modules over non-fields.

- If R is a commutative ring with 1, and M and N are left R-modules, the definition in $(\dagger)$ copies over exactly.
- If R is non-commutative with 1, then the definition in $(\dagger)$ makes sense, but the scalar multiplication in $(*)$ does *not* hold.

  In fact, we need to change the assumptions for M and N as R-modules. In particular, we need M to be a *right* R-module, and N to be a left R-module, and take the generators of type (III) for K to be defined by

  (III') $(mr,n)-(m,rn)$

  for $m \in M$, $n \in N$, and $r \in R$. This gives the tensor product $M \otimes_R N$ an abelian group structure, but does not endow it with a R-module structure.

We may now consider some simple examples computing tensor products.

Example: Let $R = \mathbb{Z}$. We will show that $\mathbb{Z}/n\mathbb{Z} \otimes_R \mathbb{Q} = 0$.

As a general strategy, in order to show that a tensor product is the zero module, it suffices to show for every simple tensor. Observe that $0 \otimes y = 0$ for any tensor product, since we may take

$$\begin{aligned} 0 \otimes y &= (0+0)\otimes y \\ &= 0\otimes y + 0 \otimes y. \end{aligned}$$

Therefore, we may write

$$[a]\otimes b = (n[a])\otimes\left(\frac{b}{n}\right)$$

$$\begin{aligned}
&= [na] \otimes \frac{b}{n} \\
&= 0 \otimes \frac{b}{n} \\
&= 0.
\end{aligned}$$

Universal Property

We may now work towards understanding one of the defining properties of tensor products in general. This requires a discussion of a weakened version of R-bilinear maps.

Definition: Let R be a ring, M a right R-module, N a left R-module, and L an abelian group written additively. A map $\varphi\colon M \times N \to L$ is called *R-balanced* if

(BM1) $\varphi(m_1 + m_2, n) = \varphi(m_1, n) + \varphi(m_2, n)$

(BM2) $\varphi(m, n_1 + n_2) = \varphi(m, n_1) + \varphi(m, n_2)$

(BM3) $\varphi(mr, n) = \varphi(m, rn)$

for all $r \in R$, $m, m_1, m_2 \in M$, and $n, n_1, n_2 \in N$.

Theorem: Let R, M, N, L be as above. Let

$$\begin{aligned}
\Omega &= \{\Phi\colon M \otimes N \to L \mid \Phi \text{ a group homomorphism}\} \\
\Delta &= \{\varphi\colon M \times N \to L \mid \varphi\ R\text{-balanced}\}.
\end{aligned}$$

Define the map $J\colon \Omega \to \Delta$ by

$$(J\Phi)(m, n) = \Phi(m \otimes n).$$

Then, J is bijective.

Proof. We have that J is injective since $J\Phi$ captures the value of Φ on simple tensors, and Φ is completely determined by its value on simple tensors since Φ is a group homomorphism, and elements of $M \otimes N$ are sums of simple tensors.

To prove surjectivity, we recall that

$$M \otimes N = \mathbb{Z}[M \times N]/\langle K \rangle.$$

Let $\varphi\colon M \times N \to L$ be an R-balanced map. By the universal property for free modules, there is a homomorphism $\widetilde{\varphi}\colon \mathbb{Z}[M \times N] \to L$ taking $(m, n) \mapsto \varphi(m, n)$.

We only need to show now that $\widetilde{\varphi}$ kills the elements of K that generate $\langle K \rangle$, but this follows from the fact that φ is R-balanced. Therefore, we get an induced map

$$\begin{aligned}
\Phi\colon M \otimes N &\to L \\
m \otimes n &\mapsto \varphi(m, n),
\end{aligned}$$

so we are done. □

Definition: Let R be a commutative ring, M, N, L left R-modules. A map $\varphi\colon M \times N \to L$ is called R-bilinear if it satisfies (BM1), (BM2), and

(BM3') $\varphi(m, rn) = \varphi(rm, n) = r\varphi(m, n)$

Theorem: If R, M, N, L are as above, then there exists a natural bijection between $\hom_R(M \otimes N, L)$ and $\hom_R(M \times N, L)$.

The proof is the same as the proof in the case of R-balanced maps, mutatis mutandis.

Proposition: Let R be a commutative ring, and M, N free left R-modules with respective bases X and Y. Then,

$M \otimes N$ is a free module with basis

$$Z = \{x \otimes y \mid x \in X, y \in Y\}.$$

Proof. We have that Z generates $M \otimes N$ as a R-module, so we only need to show that Z is linearly independent.

Let

$$v = \sum_{i=1}^{t} r_i x_i \otimes y_i.$$

Without loss of generality, we assume that $r_1 \neq 0$. It is enough to find a homomorphism $\varphi\colon M \otimes N \to R$ such that $\varphi(v) \neq 0$.

Toward this end, we construct an R-bilinear map, which we only need to specify on the basis. Define

$$\begin{aligned}
\alpha\colon M &\to R \\
x_i &\mapsto \begin{cases} 0 & x_i \neq x_1 \\ 1 & x_i = x_1 \end{cases} \\
\beta\colon N &\to R \\
y_i &\mapsto \begin{cases} 0 & y_i \neq y_1 \\ 1 & y_i = y_1 \end{cases}.
\end{aligned}$$

The map $\varphi\colon M \times N \to R$ given by $\varphi(x_i, y_i) = \alpha(x_i)\beta(y_i)$ is thus R-bilinear and induces a map on the tensor product that is nonzero at v. Thus, v is not the zero vector. □

Module Structure on Tensor Products

Thus far, we have only shown that there is a module structure on the tensor product whenever we are considering modules over commutative rings. Else, we only have an abelian group. We will specify the case when there is a module structure on the tensor product.

Definition: Let R and S be unital rings. An (S, R)-bimodule is an abelian group $(M, +)$ that is both a left S-module and right R-module satisfying the compatibility condition

$$(sm)r = s(mr)$$

for all $m \in M$, $r \in R$, and $s \in S$.

Example:

(i) If R and S are both subrings of the same ring T with $1_R = 1_S = 1_T$, then T is an (S, R) bimodule, where S acts by left-multiplication and R acts by right-multiplication.

(ii) If R is commutative, M a left R-module, then M can be considered as an (R, R)-bimodule by setting $m.r = rm$.

Proposition: Let R and S be rings, M an (S, R)-bimodule, and N a right R-module. Then, $M \otimes N$ can be endowed with a unique S-module structure by taking $s(m \otimes n) = sm \otimes n$.

Proof. Fix $s \in S$. Define the map $\varphi_s\colon M \times N \to M \otimes_R N$ by taking $\varphi_s(m, n) = sm \otimes n$.

This map is R-balanced; we will verify (BM3) for this purpose:

$$\begin{aligned}
\varphi_s(mr, n) &= s(mr) \otimes n \\
&= (sm)r \otimes n \\
&= sm \otimes rn \\
&= \varphi_s(m, rn).
\end{aligned}$$

Thus, by the universal property, there is a homomorphism of abelian groups $\overline{\varphi_s}\colon M\otimes N \to M\otimes N$ such that $\overline{\varphi_s}(m\otimes n) = sm\otimes n$. We will then define the action of s on $M\otimes N$ by taking $s.u = \overline{\varphi_s}(u)$ for any $u \in M\otimes N$. □

The most useful case for this proposition is the *extension of scalars*. If $R \subseteq S$ is a unital subring, then we may view S as an (S,R)-bimodule; for any left R-module N, we have that $S\otimes_R N$ may be endowed with the structure of an S-module. We call this module the extension of scalars of N from R to S.

Definition: Let R be a commutative ring with 1. An R-algebra is a ring A which is also an R-module such that multiplication $\mu\colon A\times A \to A$, $\mu(a,b) = ab$ is an R-bilinear map.

Example (Some R-algebras)**:** The following are R-algebras:

(i) the polynomial ring $R[x_1,\dots,x_n]$ in commuting variables;

(ii) the ring of noncommutative polynomials $R\langle x_1,\dots,x_n\rangle$;

(iii) the matrices $\operatorname{Mat}_n(R)$ of $n\times n$ matrices over R.

Theorem: Let A and B be R-algebras, where R is commutative. Then, $A\otimes B$ has a unique R-algebra structure such that

$$(a\otimes b)(c\otimes d) = ac\otimes bd.$$

Proof. See Proposition 1.4.3 in this document. □

Lemma: If $R \subseteq S$ is a unital subring, with S commutative, and M is an R-algebra, then $S\otimes_R M$ is an S-algebra.

Lemma: With the same assumptions as above, we have isomorphisms

$$\begin{aligned} S\otimes_R \operatorname{Mat}_n(R) &\cong \operatorname{Mat}_n(S)\\ S\otimes_R R[x_1,\dots,x_n] &\cong S[x_1,\dots,x_n]\\ S\otimes_R R[x]/(f(x)) &\cong S[x]/(f(x)). \end{aligned}$$

The general proof approach for these types of problems is to use the universal properties to construct an isomorphism of abelian groups, then prove that the isomorphism is compatible with the S-action and products of elements.

Finitely Generated Modules over PIDs

We will now turn our focus towards proving the structure theorem for finitely generated modules over a PID. For this, we will use the Smith Normal Form.

Smith Normal Form

Definition: Let R be a PID, and $A \in \operatorname{Mat}_n(R)$. We say A is in Smith normal form if there exists $m \leq \min(k,n)$ and elements $d_1,\dots,d_m$ such that

$$A = \begin{pmatrix} d_1 & \cdots & 0 & 0\\ 0 & \ddots & 0 & 0\\ 0 & 0 & d_m & 0\\ \vdots & \vdots & \ddots & \vdots\\ 0 & 0 & \cdots & 0 \end{pmatrix},$$

where $d_i \neq 0$ for all i and $(d_1)\supseteq(d_2)\supseteq\cdots\supseteq(d_m)$.

Theorem: If R is a PID with $A \in \operatorname{Mat}_n(R)$, then there are $U \in \operatorname{GL}_k(R)$ and $V \in \operatorname{GL}_n(R)$ such that UAV is in Smith normal form.

This decomposition is essentially unique, in that if we have two Smith normal forms

$$D = \begin{pmatrix} d_1' & \cdots & 0 & 0 \\ 0 & \ddots & 0 & 0 \\ 0 & 0 & d_m & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}$$

$$D' = \begin{pmatrix} d_1' & \cdots & 0 & 0 \\ 0 & \ddots & 0 & 0 \\ 0 & 0 & d_s' & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix},$$

then $s = m$ and d_i and d_i' are associates for each i.

To show existence, we consider the elementary row (and column) operations, viewed as elementary matrices. Given $\lambda \in R$, we consider $E_{ij}(\lambda)$ with $i < j$ to be a square matrix of size to be determined size that has 1 along the diagonal, λ in position (i, j), and 0 everywhere else. Furthermore, for any $i \neq j$, we let F_{ij} be the matrix obtained from an identity matrix of a certain size by swapping rows i and j.[I] Then, we have the following table of correspondences.

Elementary Row Operation	Interpretation as Matrix Multiplication
$R_i(A) \mapsto R_i(A) + \lambda R_j(A)$	$A \mapsto A_{ij}(\lambda)A$
$C_i(A) \mapsto C_i(A) + \lambda C_j(A)$	$A \mapsto AE_{ji}(\lambda)$
$R_i(A) \leftrightarrow R_j(A)$	$A \mapsto F_{ij}A$
$C_i(A) \leftrightarrow C_j(A)$	$A \mapsto AF_{ij}$

Thus, it suffices to show that every A can be replaced via these elementary row operations to a matrix in Smith normal form.

We will prove the special case where R is a Euclidean domain, since the primary applications we will be using are for the cases of $\mathbb{Z}$ and $F[x]$ for some field F, and both of these rings are Euclidean domains. If N is the Euclidean norm for R, then we will let $N(A) = \min_{a_{ij} \neq 0} N\left(a_{ij}\right)$ denote the norm of the matrix.

Proof of Existence. By induction, we may assume that existence holds for any matrices with smaller values of $k+n$. Starting with an arbitrary matrix with dimensions $k \times n$, we claim that we can reduce any nonzero matrix to either

(i) a matrix A' with smaller norm

(ii) a matrix of the block diagonal form

$$\begin{pmatrix} d_1 & 0 \\ 0 & B \end{pmatrix}, \tag{†}$$

where d_1 divides all the entries of B.

To see that this suffices to prove existence of Smith normal form, we start by seeing that if case 1 occurs, we apply the reduction until case 2 occurs, which necessarily happens by the well-ordering principle. Then, if A is reduced to a matrix of the block diagonal form (†), then by the induction hypothesis we may reduce B to Smith normal form using the elementary operations. We may then apply these same operations to A to yield A to have $d_2|d_3|\cdots|d_m$ and the block d_1 on the upper left hand corner, so since these elementary row

[I] One may recognize this as the matrix corresponding to the permutation $\tau = (i, j)$.

operations yield R-linear combinations of the original element, with d_1 dividing all the entries of B, we have that $d_1|d_2$.

Now, we will show that this is satisfactory. Using column and row flips, we may assume that $N(A) = N(a_{11})$. Write

$$\begin{aligned} a_{1j} &= a_{11}q_{1j} + r_{1j} \\ a_{i1} &= a_{11}q_{i1} + r_{i1} \end{aligned}$$

for all $2 \leq i, j \leq n$. Then, for any i and j, we have $N(r_{i1}) < N(a_{11})$ and $N\left(r_{1j}\right) \leq N(a_{11})$, or we have $N(r_{i1}) = 0$ or $N\left(r_{1j}\right) = 0$. By applying the operations $C_j(A) \mapsto C_j(A) - q_{1j}C_1(A)$ and $R_i(A) \mapsto R_i(A) - q_{i1}R_1(A)$, we may perform the replacements $a_{i1} \mapsto r_{i1}$ and $a_{1j} \mapsto r_{1j}$. If there is at least one such nonzero remainder, then we have $N(A') < N(A)$, and this shows case (i).

If all the remainders are 0, then we have the block diagonal form (†), and split into two subcases. If we had $a_{11}|b_{ij}$ for all $b_{ij} \in B$, then we would be done. Else, suppose there were some $s, t \geq 2$ with $a_{11} \nmid a_{st}$. In this case, we may first perform the operation $R_1(A') \mapsto R_1(A') + R_s(A')$, write $a_{st} = a_{11}q + r$ with $r \neq 0$, and then perform $C_t(A) = C_t(A) - qC_1(A)$ to find that $N(A')$ is not minimal, so we may perform the process further. This gives existence. □

Structure of Finitely-Generated Modules over PIDs

Theorem (Uniqueness of Rank)**:** Let $R \neq \{0\}$ be a commutative ring. If $R^n \cong R^m$, then $n = m$.

Proof. The theorem holds for the case of R as a field, and otherwise, pass to a quotient by a maximal ideal. □

Corollary: If M is a finitely generated free module over a commutative ring R, then if we define $\operatorname{rk}(M)$ to be the value of $n \in \mathbb{Z}_{\geq 0}$ such that $M \cong R^n$, such a value is then unique.

Theorem (Compatible Basis Theorem)**:** Let R be a PID, and M a finitely generated free module with rank n. Let N be a submodule of M. Then, N is free with rank $m \leq n$, and there is a basis $\{y_1, \dots, y_n\}$ for M and elements $(d_1) \supseteq \cdots \supseteq (d_m)$ in R such that $\{d_1y_1, \dots, d_my_m\}$ is a basis for N.

Moreover, m is unique, and $d_1, \dots, d_m$ are unique up to associates.

Proof. Let $\{x_1, \dots, x_n\}$ be a basis for M, and let N be generated by $\{u_1, \dots, u_k\}$. We may write

$$u_i = \sum_{j=1}^{n} a_{ij}x_j$$

for some $a_{ij} \in R$. Writing the system of equations as a matrix

$$\begin{pmatrix} u_1 \\ \vdots \\ u_k \end{pmatrix} = \left(a_{ij}\right)_{i,j} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix},$$

we may then put the matrix $A = \left(a_{ij}\right)_{i,j}$ in Smith normal form, giving $A = PDQ$ for some diagonal matrix $D = \operatorname{diag}_{k,n}(d_1, \dots, d_m)$, $P \in \operatorname{GL}_k(R)$, and $Q \in \operatorname{GL}_n(R)$. Write

$$\begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} = Q \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$$

$$\begin{pmatrix} v_1 \\ \vdots \\ v_k \end{pmatrix} = P^{-1} \begin{pmatrix} u_1 \\ \vdots \\ v_k \end{pmatrix}.$$

Then, $\{y_1,\dots,y_n\}$ form a basis for M, and $\{v_1,\dots,v_k\}$ generate N; since we may write

$$\begin{pmatrix} v_1 \\ \vdots \\ v_k \end{pmatrix} = D \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix},$$

it follows that $k = m$, $v_i = d_i y_i$, and we are done. Linear independence follows from the fact that R is an integral domain and that $\{y_1,\dots,y_n\}$ is a basis. □

Theorem (Classification: Invariant Factors Form)**:** Let R be a PID, and M an arbitrary finitely generated R-module. Then,

$$M \cong \left(\bigoplus_{i=1}^{\ell} R/(a_i) \right) \oplus R^s,$$

where $\ell, s \in \mathbb{Z}_{\geq 0}$, $a_1,\dots,a_\ell \in R$ are nonzero non-units, and we have $(a_1) \supseteq (a_2) \supseteq \cdots \supseteq (a_\ell)$.

Moreover, ℓ, s are unique, and $a_1,\dots,a_\ell$ are unique up to associates.

Proof. We show existence; uniqueness will follow from the discussion after the proof of the classification in elementary divisors form.

Let $\{x_1,\dots,x_n\}$ be generators for M; then, there exists a surjective R-module homomorphism $\pi\colon F = R^n \to M$ given by $(r_1,\dots,r_n) \mapsto \sum_{i=1}^n r_i x_i$. Let $N = \ker(\pi)$, so that $M \cong F/N$.

By the compatible basis theorem, there are $y_1,\dots,y_n \in F$ and $a_1,\dots,a_m \in R$ with $m \leq n$, $(a_1) \supseteq (a_2) \supseteq \cdots \supseteq (a_m) \neq 0$ such that $\{y_1,\dots,y_n\}$ are a basis for F and $\{a_1 y_1,\dots,a_m y_m\}$ are a basis for N.

By definition, we then have

$$\begin{aligned} F &= \bigoplus_{i=1}^{n} R y_i \\ N &= \bigoplus_{i=1}^{m} (a_i) y_i. \end{aligned}$$

Now, since the $(a_i)y_i$ are submodules of Ry_i, we have

$$\begin{aligned} M &\cong F/N \\ &= \bigoplus_{i=1}^{n} R y_i \Big/ \bigoplus_{i=1}^{m} (a_i) y_i \\ &= \bigoplus_{i=1}^{m} R/(a_i) \oplus R^{n-m}. \end{aligned}$$

Moreover, $(a_i) = R$ if a_i is a unit, so we may remove such terms. □

Theorem (Classification, Invariant Factors)**:** Let M be a finitely generated module over a PID. Then, there are primes $p_1,\dots,p_t \in R$ not necessarily distinct, and $d_1,\dots,d_t \in \mathbb{N}$ such that

$$M \cong \left(\bigoplus_{i=1}^{t} R/\left(p_i^{d_i}\right) \right) \oplus R^s.$$

Moreover, s, t are unique, and $p_1^{d_1},\dots,p_t^{d_t}$ are unique up to associates and permutations. We call the collection $\left\{p_1^{d_1},\dots,p_t^{d_t}\right\}$ the elementary divisors for M.

Proof. It suffices to consider the case of $M = R/(a)$, where a is a nonzero non-unit in R. Since PIDs are UFDs, we may then write

$$a = u\prod_{i=1}^{t} p_i^{d_i},$$

where u is a unit and the p_i are pairwise non-associate primes. Then, we have that $\left(p_i^{d_i}\right)$ and $\left(p_j^{d_j}\right)$ for $i \neq j$ are comaximal ideals, meaning that by the Chinese Remainder Theorem, we may write

$$R/(a) \cong \bigoplus_{i=1}^{t} R/\left(p_i^{d_i}\right).$$

□

As for uniqueness, we start by showing uniqueness of the free part, which is simpler. Observe that if we let $F = \operatorname{frac}(R)$ be the field of fractions, then

$$\begin{aligned} F \otimes_R M &= F \otimes \left(\left(\bigoplus_{i=1}^{t} R/\left(p_i^{d_i}\right)\right) \oplus R^s\right) \\ &\cong F^s, \end{aligned}$$

meaning that we may define $s := \dim(F \otimes_R M)$. Since dimensions of vector spaces are invariant, it follows that s is well-defined.

Next, we show the uniqueness of the elementary divisor decomposition. We let

$$\begin{aligned} M &\cong M_0 \\ &:= \left(\bigoplus_{i=1}^{t} R/\left(p_i^{d_i}\right)\right) \oplus R^s \end{aligned}$$

as in the elementary divisor decomposition. We let $\varphi\colon M \to M_0$ be a fixed isomorphism. Given $x, y \in R$, we will write $x \sim y$ whenever x and y are associates. For all primes p and all $d \in \mathbb{N}$, define

$$\begin{aligned} \mu_{p,d}(M_0) &= \left|\left\{i \;\middle|\; p_i^{d_i} \sim p^d\right\}\right| \\ &= |\{i \mid d_i = d, p_i \sim p\}|. \end{aligned}$$

We will show that $\mu_{p,d}$ is entirely determined by p, d, M. We will let $W = \{([p], d) \mid p \text{ is a prime}, d \in \mathbb{N}\}$.

Notice that

$$t = \sum_{([p],d)\in W} \mu_{p,d}(M_0).$$

Furthermore, we have $\varphi\left(p^d M\right) = p^d M_0$. We will examine the quotient $p^d M/p^{d+1}M$. Using the isomorphism φ and trundling through much grunt-work, we find that

$$p^d M_0/p^{d+1}M_0 \cong \left(\bigoplus_{i=1}^{t} p^d\left(R/\left(p_i^{d_i}\right)\right)/p^{d+1}\left(R/\left(p_i^{d_i}\right)\right)\right) \oplus (R/(p))^s.$$

Examining the quotients in the torsion part, we observe that, as a general rule, we have

$$\begin{aligned} z(R/I) &= \{z(r+I) \mid r \in R\} \\ &= \{zr + I \mid r \in R\} \end{aligned}$$

$$\begin{aligned}
&= ((z)+I)/I \\
&:= \{u+I \mid u \in (z)\},
\end{aligned}$$

so in our case, this yields

$$z(R/(y)) = (\gcd(z,y))/(y).$$

Therefore, returning to the quotient $p^d M_0/p^{d+1}M_0$, we observe that if p, p_i are not associate, then the corresponding quotient is zero, and if p, p_i are associates, then

$$p^d\left(R/\left(p^{d_i}\right)\right) = \left(p^{\min(d,d_i)}\right)/\left(p^{d_i}\right),$$

meaning that the corresponding quotient is isomorphic to

$$\left(p^{\min(d,d_i)}\right)/\left(p^{\min(d+1,d_i)}\right).$$

If $d \geq d_i$, then the quotient is zero, and if $d < d_i$, then the quotient is $\left(p^d\right)/\left(p^{d+1}\right) \cong R/(p)$. Since prime ideals are maximal in PIDs, we have that

$$p^d M/p^{d+1}M \cong (R/(p))^{g_{p,d}}$$

where

$$\begin{aligned}
g_{p,d} &= \dim_{R/(p)}\left(p^d M/p^{d+1}M\right) \\
&= s + |\{i \mid p_i \sim p, d_i > d\}| \\
&= s + \sum_{k \geq d+1} \mu_{p,k}(M_0).
\end{aligned}$$

Observe then that for any $d \in \mathbb{N}$, we have

$$\begin{aligned}
\mu_{p,d}(M_0) &= g_{p,d-1} - g_{p,d} \\
&= \dim_{R/(p)}\left(p^{d-1}M/p^d M\right) - \dim_{R/(p)}\left(p^d M/p^{d+1}M\right)
\end{aligned}$$

Since these quantities are uniquely determined by M, it follows that the elementary divisors decomposition of M is unique.

As for the uniqueness of the invariant factors decomposition, we write

$$M \cong \left(\bigoplus_{i=1}^{\ell} R/(a_i)\right) \oplus R^s,$$

and observe that M has the multiset of elementary divisors given by

$$E = \{\text{prime powers in the prime factorization of the } a_i\}.$$

Replacing primes by associates, we may write E as a collection of $\left\{p_i^{b_{i,j}} \;\middle|\; 1 \leq i \leq h, 1 \leq j \leq n_i\right\}$, where $p_1, \dots, p_h$ are pairwise non-associates, and $b_{i,1} \geq b_{i,2} \geq \cdots \geq b_{i,n_i}$.

Since E is unique by the uniqueness of the elementary divisors decomposition, and $(a_1) \supseteq (a_2) \supseteq \cdots \supseteq (a_\ell)$, with the a_i non-associate, we have then that

$$\begin{aligned}
a_\ell &= \prod_{j=1}^{h} p_j^{b_{j,1}} \\
a_{\ell-1} &= \prod_{j=1}^{h} p_j^{b_{j,2}} \\
&\vdots
\end{aligned}$$

so that the invariant factors are themselves unique.

Invariant Factors and Rational Canonical Form

Now that we've proven the classification of finitely generated modules over PIDs, we will use the special case of $F[x]$-modules to determine the rational canonical form.

Given a field F and a finite-dimensional vector space V over F, with $T\colon V \to V$ an F-linear transformation, we want to find a basis β of V such that $[T]_\beta$ is a simple form. The first one we will construct is the Rational Canonical Form, which exists over any field.

Note that if we let $x \cdot v = Tv$, then we may turn V into an $F[x]$-module. This is the only $F[x]$-module structure on V that extends the original vector space structure with x acting as T.

Since V is finite-dimensional over F, it is finitely generated as an F-module, so it is finitely-generated as an $F[x]$-module. Therefore, we may write

$$\begin{aligned} V &\cong \widetilde{V} \\ &:= \bigoplus_{i=1}^{n} \frac{F[x]}{(a_i(x))}, \end{aligned}$$

with $a_1(x)|a_2(x)|\cdots|a_m(x)$ nonzero non-unit elements. Note that there is no free summand since $F[x]$ itself is infinite-dimensional as a vector space over F, so since V is assumed to be finite-dimensional, it follows that there is no free summand.

We may safely assume that the a_i are monic, so that the a_i are unique. We call them the invariant factors of this linear transformation.

We let $\varphi\colon V \to \widetilde{V}$ be an isomorphism of $F[x]$-modules, which gives rise to the following commutative diagram.

$$\begin{tikzcd} V \arrow[r, "T"] \arrow[d, "\varphi"'] & V \arrow[d, "\varphi"] \\ \widetilde{V} \arrow[r, "\varphi\circ T\circ\varphi^{-1}"'] & \widetilde{V} \end{tikzcd}$$

Note that if β is a basis of V, then $\varphi(\beta)$ is a basis for $\varphi(V)$, and we have $[T]_\beta = \left[\varphi \circ T \circ \varphi^{-1}\right]_{\varphi(\beta)}$, by the definition of the basis.

Note that if $w \in \widetilde{V}$, then the action of x on w is given by

$$\begin{aligned} x \cdot w &= x \cdot \varphi\left(\varphi^{-1}(w)\right) \\ &= \varphi\left(x \cdot \varphi^{-1}(w)\right) \\ &= \varphi\left(T \circ \varphi^{-1}(w)\right) \\ &= \left(\varphi \circ T \circ \varphi^{-1}\right)(w). \end{aligned}$$

Therefore, we may assume that in fact $V = \widetilde{V}$. Writing

$$V_i = F[x]/(a_i(x)),$$

then we have

$$V = \bigoplus_{i=1}^{m} V_i,$$

and if β_i is a basis for V_i, then

$$[T]_\beta = \operatorname{diag}\left(\left[T|_{V_1}\right]_{\beta_1}, \dots, \left[T|_{V_m}\right]_{\beta_m}\right).$$

Therefore, it suffices to consider the case of one invariant factor,

$$V = F[x]/(a(x)),$$

where $a(x)$ is a nonconstant monic polynomial of degree $d > 0$. Write

$$a(x) = x^d + \sum_{i=0}^{d-1} a_i x^i.$$

Then, for any $p(x) \in F[x]$, we define $\overline{p(x)} = p(x) + (a(x))$, and observe that $\left\{\overline{1}, \overline{x}, \dots, \overline{x}^{d-1}\right\} = \beta$ is a basis for V, and we have $T\left(\overline{x}^i\right) = \overline{x}^{i+1}$, and we have the matrix form

$$[T]_\beta = \begin{pmatrix} 0 & 0 & \cdots & 0 & -a_0 \\ 1 & \ddots & \cdots & \vdots & -a_1 \\ 0 & 1 & \ddots & \vdots & \vdots \\ 0 & 0 & \ddots & \vdots & -a_{d-2} \\ 0 & 0 & \cdots & 1 & -a_{d-1} \end{pmatrix}$$

This matrix is known as the *companion matrix* of the polynomial $a(x)$, and is denoted $C_{a(x)}$. In the general case, we have the block-diagonal matrix

$$[T]_\beta = \operatorname{diag}\left(C_{a_1(x)}, \dots, C_{a_m(x)}\right).$$

The matrix of this form is known as the *rational canonical form* of the linear transformation T.

Theorem: Let V be a finite-dimensional vector space over F, and let $T\colon V \to V$. Then, there exists a basis β of V and non-constant monic polynomials $a_1, \dots, a_m(x) \in F[x]$ with $a_1(x) | \cdots | a_m(x)$ such that

$$[T]_\beta = \operatorname{diag}\left(C_{a_1(x)}, \dots, C_{a_m(x)}\right)$$

is in block-diagonal form, where $C_{a_i(x)}$ is the companion matrix of $p(x)$.

Moreover, the number m and the polynomials $a_i(x)$ are unique. The polynomials $a_i(x)$ are called the invariant factors of T.

Corollary: Let K/F be a field extension. Suppose $A, B \in \operatorname{Mat}_n(F)$. Then, A and B are F-similar if and only if A and B are K-similar.

Proof. The forward direction is clear.

In the reverse direction, if A and B are not F-similar, then the RCF of A is not equal to the RCF of B over F, so they are not equal over K, so A and B are not similar over K. □

Theorem: If A is a matrix, then the following Smith Normal Form computes the RCF of A:

$$\operatorname{SNF}(xI - A) = \operatorname{diag}(1, \dots, 1, a_1(x), \dots, a_m(x)),$$

where there are $n - m$ instances of 1.

Proof. First, it can be shown that if A is the companion matrix of $a(x) \in F[x]$, then

$$\operatorname{SNF}(xI - A) = \operatorname{diag}(1, \dots, 1, a(x)).$$

Then, if A is an arbitrary matrix in rational canonical form, then we reduce to the first case by restricting the row and column operations to each block.

Finally, if A is a general matrix, then if we set $B = \operatorname{RCF}(A)$, then A and B have the same invariant factors, so $\operatorname{SNF}(xI - B)$ is in the desired form. We observe then that

$$\operatorname{SNF}\left(P(xI - B)P^{-1}\right) = \operatorname{SNF}(xI - A),$$

so we are done. □

Now, we can relate the RCF to the minimal and characteristic polynomials of linear transformations.

Definition: Let F be a field, $n \in \mathbb{N}$, $T\colon V \to V$ a linear map between n-dimensional vector spaces. The polynomial $\chi_T(x) := \det(xI - T)$ is called the *characteristic polynomial* of T.

Definition: Let F, V, T be as above. The *minimal polynomial* of T, denoted $\mu_T(x)$, is the unique monic polynomial of minimal degree such that $\mu_T(T) = 0$.

For existence, observe that $\left\{1, T, T^2, \dots, T^{n^2}\right\}$ are linearly dependent in the space $\operatorname{end}_F(V)$, so there is some dependence relation among them. We may then take the dependence relation with smallest degree.

For uniqueness, observe that if there are two such minimal polynomials, then their difference is a polynomial of strictly smaller degree, so it must be equal to 0.

Definition: Let $V = V_T$ be the vector space V considered as an $F[x]$-module, where we let $x \cdot v = Tv$. Define

$$\begin{aligned} I_T &:= \operatorname{ann}_{F[x]}(V_T) \\ &= \{p(x) \in F[x] \mid p(x) \cdot v = 0 \text{ for all } v \in V\} \\ &= \{p(x) \in F[x] \mid p(T) = 0\}. \end{aligned}$$

Then, I_T is a nonzero ideal in $F[x]$, so I_T is generated by a single element (since $F[x]$ is a PID); we may also define μ_T to be the unique monic polynomial that generates I_T.

Theorem: Let $T\colon V \to V$ be a linear map, and let $a_1(x), \dots, a_m(x)$ be its invariant factors. Then,

$$\begin{aligned} \chi_T(x) &= \prod_{k=1}^{m} a_k(x) \\ \mu_T(x) &= a_m(x). \end{aligned}$$

Proof. Choose a basis β of V, and let $A = [T]_\beta$. Then, there is some unit $u \in (F[x])^\times$ such that

$$\begin{aligned} \chi_T(x) &= \chi_A(x) \\ &= \det(xI - A) \\ &= u \det(\operatorname{SNF}(xI - A)) \\ &= u \prod_{k=1}^{m} a_k(x), \end{aligned}$$

but since $\chi_T(x)$ is monic, $u = 1$.

By the invariant factors decomposition of V, we have

$$V \cong \bigoplus_{i=1}^{m} F[x]/(a_i(x)),$$

with $(a_1(x)) \supseteq \cdots \supseteq (a_m(x))$. If $p(x) \in F[x]$, then $p(x) \in \operatorname{ann}(V_T)$ if and only if $p(x) \in \operatorname{ann}(F[x]/(a_i(x)))$ for each $i = 1, \dots, m$, which holds if and only if $p(x) \in (a_i(x))$ for each i. Thus, $a_m(x)$ is a generator for $(\mu_T(x))$, so since $a_m(x)$ is also monic, it follows that $a_m(x) = \mu_T(x)$. □

Corollary (Cayley–Hamilton Theorem)**:** If $T\colon V \to V$ is a linear transformation between finite-dimensional vector spaces, then $\mu_T(x) | \chi_T(x)$.

Furthermore, there is some exponent k such that $\chi_T(x) | (\mu_T(x))^k$.

Elementary Divisors and Jordan Canonical Form

Let F be an algebraically closed field, V a finite-dimensional vector space over F, and $T\colon V \to V$ a linear transformation.

Turn V into an $F[x]$-module as in the construction of RCF; using the classification in elementary divisors, we may take

$$V \cong \bigoplus_{k=1}^{m} F[x]/\left(p_k(x)^{d_k}\right),$$

where each $p_k(x)$ is prime and $d_k \in \mathbb{N}$. Since F is algebraically closed, the only irreducible factors in $F[x]$ are the linear factors, so we may write

$$V \cong \bigoplus_{k=1}^{m} F[x]/\left((x-\lambda_k)^{d_k}\right),$$

where λ_k may be repeated.

It thus suffices to consider the case

$$V \cong F[x]/(x-\lambda)^d,$$

and attempt to construct a canonical form for the action of T on V as multiplication by x.

For any $p(x)$, let $\overline{p(x)} = p(x) + \left((x-\lambda)^d\right)$. It can be verified that

$$\beta = \left\{\overline{(x-\lambda)}^{d-1}, \dots, \overline{(x-\lambda)}, \overline{1}\right\}$$

forms a basis for V over F.

Consider the transformation $S = T - \lambda I$, so that $T = S + \lambda I$. Observe that S acts on V by multiplication by $x - \lambda$. Then, we have that $[S]_\beta$ consists of the matrix exclusively of 1 on the superdiagonal,

$$[S]_\beta = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 1 \\ 0 & 0 & 0 & \cdots & 0 & 0 \end{pmatrix}.$$

Thus, we find that

$$[T]_\beta = \begin{pmatrix} \lambda & 1 & 0 & \cdots & 0 & 0 \\ 0 & \lambda & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & 0 \\ 0 & 0 & 0 & \cdots & \lambda & 1 \\ 0 & 0 & 0 & \cdots & 0 & \lambda \end{pmatrix}$$

We call a matrix of this form the *Jordan block* with eigenvalue λ and size d, denoted $J_d(\lambda)$. Combining this work gives us the following theorem, with proof similar to the proof of the rational canonical form, replacing the use of the invariant factor decomposition with the elementary divisor decomposition.

Theorem (Jordan Canonical Form)**:** Let T be a linear transformation on V, where V is a finite-dimensional vector space over an algebraically closed field F.

(i) There is a basis β such that

$$\begin{aligned}[T]_\beta &= \bigoplus_{k=1}^{m} J_{d_k}(\lambda_k)\\ &= \operatorname{diag}\Big(J_{d_1}(\lambda_1),\dots,J_{d_m}(\lambda_m)\Big).\end{aligned}$$

(ii) This arrangement is unique up to the rearranging of the blocks.

There are two immediate corollaries.

Corollary: If $T\colon V \to V$ is a linear map between finite dimensional vector spaces over an algebraically closed field F, then the following hold:

(a) $\chi_T(x) = \prod_{\lambda\in\sigma(T)} (x-\lambda)^{n_\lambda}$, where n_λ is the sum of the sizes of all the Jordan blocks of T with eigenvalue λ;

(b) $\mu_T(x) = \prod_{\lambda\in\sigma(T)} (x-\lambda)^{m_\lambda}$, where m_λ is the size of the largest Jordan block with eigenvalue λ.

Corollary: Letting T be as above, the following are equivalent:

(i) T is diagonalizable, in that V has a basis of eigenvectors for T;

(ii) all the blocks in the Jordan canonical form of T have size 1;

(iii) $\mu_T(x)$ is a product of distinct linear factors.

Now, we will discuss a little bit on how to compute Jordan canonical forms by using the ranks of certain linear transformations. For any λ, we let $b_d(\lambda)$ denote the number of times the block $J_d(\lambda)$ appears in the Jordan canonical form of T. Then, the following holds.

Theorem:

$$\sum_{d\geq n} b_d(\lambda) = \operatorname{rk}\Big((T-\lambda I)^{n-1}\Big) - \operatorname{rk}((T-\lambda I)^n),$$

where we let $S^0 = I$.

Proof. We start with a general observation. If W is a finite-dimensional vector space, and $S\colon W \to W$ is a linear map, where

$$W = \bigoplus_{k=1}^{m} W_k$$

is a decomposition into S-invariant subspaces, then

$$\operatorname{rk}(S) = \sum_{k=1}^{m} \operatorname{rk}\Big(S|_{W_k}\Big).$$

Let β be a Jordan basis for T, and write

$$[T]_\beta = \bigoplus_{k=1}^{m} J_{d_k}(\lambda_k).$$

Let β_k be a subset of β corresponding to $J_{d_k}(\lambda_k)$, and set $V_k = \operatorname{Span}(\beta_k)$. Then, we have

$$V = \bigoplus_{k=1}^{m} V_k,$$

where each V_k is T-invariant, meaning each V_k is invariant under operators of the form $(T-\lambda I)^n$.

Thus, we have

$$\begin{aligned}\operatorname{rk}\left((T-\lambda I)^{n-1}\right)-\operatorname{rk}((T-\lambda I)^n) &= \sum_{k=1}^{m}\operatorname{rk}\left((T-\lambda I)^{n-1}|_{V_k}\right)-\operatorname{rk}\left((T-\lambda I)^{n}|_{V_k}\right)\\ &= \sum_{k=1}^{m}\operatorname{rk}\left(\left(J_{d_k}(\lambda_k)-\lambda I\right)^{n-1}\right)-\operatorname{rk}\left(\left(J_{d_k}(\lambda_k)-\lambda I\right)^{n}\right), \qquad (*)\end{aligned}$$

meaning that we only need to understand the ranks $\operatorname{rk}(J_d(\mu)^s)$, where $\mu \in F$, $d \in \mathbb{N}$, and $s \geq 0$. Observe that if $\mu \neq 0$, then $J_d(\mu)$ is invertible, so all its powers are full rank. Thus, if $\lambda_k \neq \lambda$, then all the summands in $(*)$ are zero.

Concerning $J_d(0)^s$, we observe that it is an $s\times s$ matrix with 1s on the superdiagonal and 0 everywhere else, so it follows that $J_d(0)^s$ takes $e_1,\dots,e_s$ to 0 and $e_{s+k} \mapsto e_k$ for $1 \leq k \leq d-s$, and if $s \geq d$ then $J_d(0)^s = 0$. Thus, the rank of $J_d(0)^s$ is given by $d-s$ if $s \leq d$ and 0 if $s \geq d$.

In particular, this means

$$\operatorname{rk}\left(J_d(0)^{n-1}\right)-\operatorname{rk}(J_d(0)^n) = \begin{cases}1 & d \geq n\\ 0 & d < n\end{cases}.$$

Furthermore, if $\lambda \notin \sigma(T)$, then $(*)$ is 0. Meanwhile, if $\lambda \in \sigma(T)$, then the Jordan block $J_{d_k}(\lambda_k)$ contributes 1 to $(*)$ if and only if $\lambda_k = \lambda$ and $d_k \geq n$. This gives our desired result. □

Representation Theory

The primary idea behind representation theory is that if we have a group G, and some class of groups $\mathcal{C}$, we want to understand homomorphisms from G to elements of $\mathcal{C}$. The primary focuses will be:

- $\operatorname{GL}(V)$, where V is a vector space over F, not necessarily finite-dimensional;
- $\operatorname{GL}_n(F)$, where F is a field and $n \in \mathbb{N}$ is some natural number.

Definition: Let G be a group.

- A linear representation of G is a pair (ρ, V), where V is a vector space over some field F, and $\rho\colon G \to \operatorname{GL}(V)$ is a homomorphism.
- A matrix representation of G is a homomorphism $\rho\colon G \to \operatorname{GL}_n(F)$, where F is some field and $n \in \mathbb{N}$ is some natural number.

Definition: Let (ρ_1, V_1) and (ρ_2, V_2) be two linear representations of G. We say $\rho_1 \cong \rho_2$ if there exists an isomorphism $T\colon V_2 \to V_1$ such that

$$\rho_2(g) = T^{-1}\rho_1(g)T$$

for all $g \in G$. We say T *intertwines* the representations ρ_1 and ρ_2.

Example:

(i) We start by discussing representation theory of cyclic groups.

- Suppose $G = \langle t\rangle \cong \mathbb{Z}$. Fixing $n \in \mathbb{N}$, and F a field, then for any $A \in \operatorname{GL}_n(F)$, we may find $\rho_A\colon G \to \operatorname{GL}_n(F)$ given by $t^k \mapsto A^k$.
- Suppose $G = \left\langle t | t^d = 1\right\rangle \cong \mathbb{Z}/d\mathbb{Z}$. Then, it is still the case that any representation is of the form ρ_A, for some $A \in \operatorname{GL}_n(F)$, defined by the same map as in the infinite cyclic case. Yet, there is not necessarily one for every A, since we need $o(A)$ to be well-defined and to divide d.

(ii) Suppose G acts on the set X. Then, for any field F, we may define the free vector space $F[X]$ to be a vector space over F with basis X. Define a linear representation $(\rho, F[X])$ by

$$\rho(g)\left(\sum_{x\in X}\alpha_x x\right) = \sum_{x\in X}\alpha_x(g\cdot x).$$

In the case that $X = G$ and G acts on G by left-multiplication, then $F[G]$ is known as the group algebra of G, and $(\lambda, F[G])$ is given by

$$\begin{aligned}\lambda(h)\left(\sum_{g\in G}\alpha_g g\right) &= \sum_{g\in G}\alpha_g(hg)\\ &= \sum_{g\in G}\alpha_{h^{-1}g}g.\end{aligned}$$

This is known as the *left-regular representation*.

(iii) We can consider representations as $F[G]$-modules, akin to how vector spaces can be viewed as $F[x]$-modules where x acts as an endomorphism.

We have a natural bijection between linear representations of a group G of the form (ρ, V) and $F[G]$-modules extending the F-vector space structure on V by defining

$$\begin{aligned}(\lambda 1).v &= \lambda v\\ \left(\sum_{\alpha} f_\alpha g_\alpha\right)v &= \sum_{\alpha} f_\alpha \rho(g_\alpha)v\end{aligned}$$

Moreover, the representations (ρ_1, V_1) and (ρ_2, V_2) are isomorphic if and only if their corresponding $F[G]$-modules are isomorphic.

(iv) Given representation ρ_1 and ρ_2 defined on F-vector spaces V_1 and V_2, we can construct other representations.

- The direct sum $\rho_1 \oplus \rho_2$ is the representation on $V_1 \oplus V_2$ given by

$$(\rho_1 \oplus \rho_2)(g)(v_1, v_2) = (\rho_1(g)v_1, \rho_2(g)v_2).$$

If V_1 and V_2 are finite-dimensional and we equip them with bases to convert ρ_1 and ρ_2 into matrix representations, then $(\rho_1 \oplus \rho_2)(g) = \operatorname{diag}(\rho_1(g), \rho_2(g))$ in block-diagonal form.

- The tensor product $\rho_1 \otimes \rho_2$ acting on $V_1 \otimes V_2$ is given by the action on simple tensors

$$(\rho_1 \otimes \rho_2)(g)(v_1 \otimes v_2) = \rho_1(g)v_1 \otimes \rho_2(g)v_2.$$

If V_1 and V_2 are finite-dimensional, then the corresponding matrix representations have their tensor product given by the *Kronecker product* of their corresponding matrices on the space

$$\mathrm{GL}_n(\mathrm{GL}_m(F)) \cong \mathrm{GL}_{mn}(F).$$

Subrepresentations, Irreducible Representations, and Completely Reducible Representations

Definition: A subrepresentation of a linear representation (ρ, V) is a subspace $W \leq V$ such that $(\rho|_W, W)$ is a linear representation.

Equivalently, a subrepresentation is a submodule of the corresponding $F[G]$-module structure on V.

Definition: Let R be a ring with 1, and let M be a left R-module.

(i) We say M is irreducible (or simple) if M is nonzero and the only submodules of M are $\{0\}$ and M.

(ii) We say M is completely reducible (or semisimple) if M can be written as

$$M = \bigoplus_{i \in I} M_i,$$

where each M_i is an irreducible submodule.

We say a representation (ρ, V) is irreducible or completely reducible if the corresponding $F[G]$-module structure has the same property.

Ideally, if G is some (finite) group, we would like all the representations of G to be completely reducible, so we would only need to study irreducible representations of groups.

We note that many modules are not completely reducible. For instance, $\mathbb{Z}$, viewed as a module over $\mathbb{Z}$, has no irreducible submodules,[II] but also $\mathbb{Z}$ itself is not an irreducible submodule.

Now, if (ρ, V) is a group representation, and $\dim(V) < \infty$ with $V \neq 0$, then V does have an irreducible subrepresentation (by the well-ordering principle), but it may fail to be completely reducible. For instance, if $G = \mathbb{Z}$, F is any field, and $V = F^2$, with the representation ρ given by

$$\rho(g) = \begin{pmatrix} 1 & g \\ 0 & 1 \end{pmatrix},$$

then we have that any proper nonzero subrepresentation of V must have dimension 1, meaning that we would have $V \cong W_1 \oplus W_2$ with each of W_1 and W_2 G-invariant. This would imply that W_1 and W_2 are the spans of eigenvectors of $\rho(g)$ for all $g \in G$, but $\rho(1)$ only has one eigenvector. Similarly, if F is a field of characteristic p, and $G = \mathbb{Z}/p\mathbb{Z}$, then we would find that (ρ, V) is not completely reducible as before.

Later, we will prove a very convenient way to show certain representations of finite groups are completely reducible.

Theorem (Maschke's Theorem)**:** If G is a finite group, F a field of characteristic q with $q \nmid |G|$ or $q = 0$, and V is a vector space over F, then any representation (ρ, V) of G is completely reducible.

Theorem (Complete Reducibility Criterion)**:** Let M be a left R-module. The following are equivalent.

(i) We may decompose $M = \bigoplus_{i \in I} M_i$, where each M_i is an irreducible submodule.

(ii) We may decompose $M = \sum_{i \in I} M_i$, where each M_i is an irreducible submodule.

(iii) For any submodule N of M, there is a submodule P such that $M = N \oplus P$. (Complement property.)

Lemma: If M is an R-module satisfying (iii), then any nonzero submodule of M contains an irreducible submodule.

Proof. We start by claiming that if $M = P \oplus Q$, then if N is any submodule containing P, we have $N = P \oplus (N \cap Q)$. For this, observe that $M/P \cong Q$, so since N contains P, N/P is a submodule of M/P with $N/P \cong N \cap Q$, meaning that $N \cong P \oplus (N \cap Q)$ via the short exact sequence $0 \to P \to N \to N \cap Q \to 0$. In particular, if M has the complement property, then so too does any submodule of M.

Now, if M has the complement property, and $N \leq M$ is a submodule with $x \in N$, we have that the submodule $L = \langle x \rangle$ generated by x also has the complement property. Any submodule of $\langle x \rangle$ can then be viewed as a submodule of R viewed as an R-module; i.e., any submodule of $\langle x \rangle$ can be viewed as an ideal of R. We see that R has a maximal ideal, P, with corresponding submodule $K = Px \leq L$. This submodule does not contain x as P does not contain 1. Since L has the complement property, $L = K \oplus B$ for some submodule $B \leq L$. Then, $L/K \cong B$, and since $L/K \cong R/P$, and R/P is a field, we have that R/P has no R-submodules other than 0 and the whole space, meaning that B has no submodules. Since $x \notin K$, we have $x \in B$, meaning that B is an irreducible submodule of L that is a submodule of N, so every submodule of M has an ir-

[II] The ring $\mathbb{Z}$ is not an Artinian ring, as we can take a descending chain $(a) \supsetneq (2a) \supsetneq (3a) \supsetneq \cdots$ for some $a \neq 1$ without any stabilization point.

reducible submodule. □

Proof of Complete Reducibility Criterion. We start by showing that (ii) implies (i). Suppose $M = \sum_{i\in I} M_i$. By Zorn's Lemma, there is a maximal subset J of I such that

$$\begin{aligned} N &= \sum_{j\in J} M_j \\ &= \bigoplus_{j\in J} M_j \end{aligned}$$

is a direct sum. We will show that $M_i \subseteq N$ for all $i \in I$. If $i \in J$, then $M_i \subseteq N$, while if $i \notin J$, then $M_i \cap N \subseteq M_i$, meaning either $M_i \cap N = 0$ or $M_i \cap N = M_i$. If it is the former, then $N + M_i$ directly contradicts maximality, so we have the latter, and thus (i) holds.

Now, suppose (iii) holds. Using the lemma, we may consider the submodule N consisting of the sum of all irreducible submodules of M. Since M has the complement property, it follows that there is a submodule P such that $M = N \oplus P$. We will show that $M = N$. If $P \neq 0$, then P has an irreducible submodule from the above lemma, meaning that the sum is not direct, meaning that M is the sum of all its irreducible submodules.

Finally, we show (i) implies (iii). Letting $M = \bigoplus_{i\in I} M_i$, and N be any submodule, then by Zorn's Lemma, there is a maximal subset J such that $P = \bigoplus_{i\in J} M_i$ has trivial intersection with N. Let $M' = N \oplus P$, which is a direct sum by construction. We will show that $M_i \subseteq M'$ for all $i \in I$. If $i \in J$, then $M_i \subseteq P \subseteq M'$, and if $i \notin J$, then since J is maximal, $N \cap (P \oplus M_i) \neq 0$, so there exists $0 \neq n \in M$ such that $n = p + m$ for some $p \in P$ and $m \in M_i$. We note that m has to be nonzero since otherwise N would have nontrivial intersection with P.

Hence, we have that $0 \neq m = n - p \in N \oplus P = M'$, meaning that $M_i \cap M' \neq 0$, so $M' \cap M_i = M_i$ by irreducibility, so $M_i \subseteq M'$. □

Unitary and One-Dimensional Representations

Definition: Let V be a (finite-dimensional) vector space over $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. Then, an inner product $\langle\cdot,\cdot\rangle\colon V \times V \to F$ is a map satisfying

(i) $\langle v, v\rangle \geq 0$ for all $v \in V$, with equality if and only if $v = 0$;

(ii) $\langle \alpha v_1 + \beta v_2, w\rangle = \alpha\langle v_1, w\rangle + \beta\langle v_2, w\rangle$;

(iii) $\langle v, \alpha w_1 + \beta w_2\rangle = \overline{\alpha}\langle v, w_1\rangle + \overline{\beta}\langle v, w_2\rangle$;

(iv) $\langle v, w\rangle = \overline{\langle w, v\rangle}$.

We call a vector space equipped with an inner product an inner product space.

Over $\mathbb{R}$, the inner product is a symmetric positive definite bilinear form, while over $\mathbb{C}$, the inner product is a conjugate-symmetric positive definite sesquilinear form.

Definition: A representation (ρ, V) on a finite-dimensional $\mathbb{C}$-inner product space is known as a *unitary representation* if, for all $g \in G$ and all $v, w \in V$, we have

$$\langle v, w\rangle = \langle \rho(g)v, \rho(g)w\rangle.$$

Any complex representation (ρ, V) of G is called *unitarizable* if there is an inner product on V such that (ρ, V) is unitary with respect to this inner product.

Lemma: If (ρ, V) is a unitary representation, then for any subrepresentation W of V, its orthogonal complement $W^{\perp}$ is also a subrepresentation.

Note that this means any unitary representation is completely reducible.

Theorem: If G is a finite group, and (ρ, V) is a finite-dimensional complex representation of G, then (ρ, V) is unitarizable.

Proof. Choose an inner product $\langle \cdot, \cdot \rangle_0$ on V, which can be constructed by defining an isomorphism $\varphi\colon V \to \mathbb{C}^n$ and taking

$$\langle v, w \rangle_0 = \langle \varphi(v), \varphi(w) \rangle_{\mathbb{C}^n}.$$

We construct a new inner product on V by taking

$$\langle v, w \rangle = \frac{1}{|G|} \sum_{g \in G} \langle \rho(g)v, \rho(g)w \rangle_0.$$

Then, $\rho(g)$ is a unitary map for any $g \in G$ with respect to this inner product. □

In particular, since any complex representation is equivalent to a unitary representation, and any unitary representation is a direct sum of irreducible representations, it suffices to study irreducible representations.

The most important theorem in this realm is Schur's Lemma, which allows us to understand the irreducible representations of abelian groups very easily.

Theorem (Schur's Lemma (Irreducible Modules))**:** Let R be a ring, and let M be an irreducible (left) R-module. Then, $\operatorname{end}_R(M)$, the ring of endomorphisms of M, is a division ring.

Theorem (Schur's Lemma (Group Representations))**:** Let F be an algebraically closed field, V a finite-dimensional vector space over F, and (ρ, V) an irreducible representation of G.

Then, if $g \in Z(G)$, we have $\rho(g) = \lambda I$ for some $\lambda \in F$.

Corollary: Any irreducible representation of an abelian group on a finite-dimensional complex vector space is 1-dimensional.

Note that the assumption of finite-dimensionality is required. If $V = F(t)$ is the space of rational functions over F in one variable (as a vector space over F), and $G = F(t)^\times$, then multiplication of elements of V by elements of G gives a representation of G that is not one-dimensional and is not able to be reduced to one dimension.

To prove the following proposition, we note that if H and K are any groups, and A is any abelian group, then there is a natural bijection

$$\hom(H \times K, A) \to \hom(H, A) \times \hom(K, A)$$

given by applying the maps.

Proposition: Let G be a finite abelian group. Then,

$$\left|\hom(G, \mathbb{C}^\times)\right| = |G|.$$

Proof. Writing out

$$G = \mathbb{Z}/p_1^{d_1}\mathbb{Z} \times \cdots \times \mathbb{Z}/p_k^{d_k}\mathbb{Z},$$

we observe that it suffices to count the number of homomorphisms from $\mathbb{Z}/n\mathbb{Z} \to \mathbb{C}^\times$. Writing $\mathbb{Z}/n\mathbb{Z} = \langle g | g^n = 1 \rangle$, we have that any homomorphism $\rho\colon G \to \mathbb{C}^\times$ is completely determined by $\rho(g)$, and the only valid choices for $\rho(g)$ are the nth roots of unity, of which there are n. □

In the case of non-abelian groups, we recall that the abelianization of G is given by

$$G^{\mathrm{ab}} = G/[G, G].$$

Informally, $G/[G, G]$ is the largest abelian quotient of G. That is, if $N \trianglelefteq G$ is such that G/N is abelian, then $[G, G] \leq N$.

Formally, G^{ab} admits the following universal property: if A is any abelian group and $\varphi\colon G \to A$ is a homomorphism, then there is a unique homomorphism $\overline{\varphi}\colon G^{\mathrm{ab}} \to A$ such that $\overline{\varphi} \circ \pi = \varphi$.

This is to say that $\hom\left(G^{\text{ab}}, A\right) \mapsto \hom(G, A)$, meaning that

$$\left|\hom(G, \mathbb{C}^{\times})\right| = \left|\hom\left(G^{\text{ab}}, \mathbb{C}^{\times}\right)\right|.$$

Character Theory

From now on, we will assume that all reps are finite-dimensional, unless stated otherwise.

Definition: Let (ρ, V) be a representation of a group G. The *character* of ρ is the function

$$\begin{aligned} \chi_\rho \colon G &\to F \\ g &\mapsto \operatorname{Tr}(\rho(g)). \end{aligned}$$

Observe that χ_ρ is a class function — that is, since $\operatorname{Tr}(AB) = \operatorname{Tr}(BA)$ for any $A, B \in \operatorname{Mat}_n(F)$,[III] it follows that for any $P \in \operatorname{Mat}_n(F)$, we have $\operatorname{Tr}\left(PAP^{-1}\right) = \operatorname{Tr}(A)$. Furthermore, equivalent representations have the same character for the same reason.

Lemma: If ρ_1, ρ_2 are representations of G over the same field, then the following hold:

(i) $\chi_{\rho_1 \oplus \rho_2} = \chi_{\rho_1} + \chi_{\rho_2}$;

(ii) $\chi_{\rho_1 \otimes \rho_2} = \chi_{\rho_1} \chi_{\rho_2}$.

Our goal is to be able to understand the character of irreducible representations if we are examining complex representations of some finite group. We will call a character of a representation of G a character of G, and an *irreducible* character is the character of some irreducible representation of G.

If G is a finite group, and $C_1, \dots, C_k$ are the conjugacy classes of G, and $\chi_1, \chi_2, \dots$ are (isomorphism classes of) irreducible characters of G, then the *character table* is the matrix whose columns are labeled by the conjugacy classes $C_1, \dots, C_k$ and rows are labeled by the irreducible characters, where the (i, j) entry is given by $\chi_i\left(g_j\right)$, where $g_1, \dots, g_k$ are representatives of the conjugacy classes.

	C_1	C_2	$\cdots$	C_k
χ_1	$\chi_1(g_1)$	$\chi_1(g_2)$	$\cdots$	$\chi_1(g_k)$
χ_2	$\chi_2(g_1)$	$\chi_2(g_2)$	$\cdots$	$\chi_2(g_k)$
χ_3	$\chi_3(g_1)$	$\chi_3(g_2)$	$\cdots$	$\chi_3(g_k)$
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$

Soon, we will show that this is in fact a square table (i.e., the number of irreducible characters is equal to the number of conjugacy classes). For now, we will take this on faith, and work through some examples.

Examples of Characters and Character Tables

Let $G = \left\langle a, b \middle| a^2, b^2, aba^{-1}b^{-1} \right\rangle \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ be the Klein 4-group. All conjugacy classes have size 1, so G has exactly four 1-dimensional representations. The character table is thus given below.

	e	a	b	ab
χ_1	1	1	1	1
χ_2	1	1	-1	-1
χ_3	1	-1	1	-1
χ_4	1	-1	-1	1

Now, one of the most important groups is the symmetric group. If $G = S_3$, then the conjugacy classes of G are defined by

$$\begin{aligned} K_1 &= \{e\} \\ K_2 &= \{(1,2), (1,3), (2,3)\} \end{aligned}$$

[III] It is often said that the trace is tracial.

$$K_3 = \{(1,2,3),(1,3,2)\}.$$

We begin by describing three non-equivalent irreducible complex representations of S_3.

Observe that $[S_n, S_n] = A_n$ for all $n \geq 3$, meaning that there are two one-dimensional representations. These are as follows:

- the trivial representation, $\rho_t \colon S_3 \to \mathbb{C}^\times$ taking $\sigma \mapsto 1$;
- the sign representation, $\rho_{\text{sgn}} \colon S_3 \to \mathbb{C}^\times$ taking $\sigma \mapsto \text{sgn}(\sigma)$.

Now, consider the *defining representation*, given by

$$\rho_{\text{def}} \colon S_3 \to \text{GL}_3(\mathbb{C})$$

given by mapping a permutation σ to the matrix taking the standard basis vectors $e_k \mapsto e_{\sigma(k)}$. Note that this is not irreducible, as the subspace

$$U = \mathbb{C}(e_1 + e_2 + e_3)$$

is invariant under $\rho_{\text{def}}(S_3)$. The orthogonal complement to this subspace is given by the subspace

$$W = \{\lambda_1 e_1 + \lambda_2 e_2 + \lambda_3 e_3 \mid \lambda_1 + \lambda_2 + \lambda_3 = 0\},$$

which is also invariant, and is also irreducible. The restriction of ρ_{def} to the subspace W is known as the *standard representation* of S_3, and is a two-dimensional representation denoted ρ_{std}.

To find the character χ_{std}, we observe that

$$\rho_{\text{def}} = \rho_{\text{std}} \oplus \rho_t$$

following from the decomposition $\mathbb{C}^3 = U \oplus W$, meaning that

$$\chi_{\text{std}} = \chi_{\text{def}} - 1.$$

This gives us the following character table.

	e	$(1,2)$	$(1,2,3)$
χ_t	1	1	1
χ_{sgn}	1	-1	1
χ_{std}	2	0	-1

Structure of Character Tables

The shape of character tables is described below.

Theorem (Shape of the Character Table)**:** Let G be a finite group, and let r be the number of conjugacy classes of G. Then, there are exactly r equivalence classes of irreducible complex representations for G. That is, the character table is a square matrix.

Letting $\chi_1, \dots, \chi_r$ denote the characters of these irreducible representations, then the following orthogonality relations hold.

Theorem (Orthogonality Relations)**:**

- **First Orthogonality Relation:** given characters χ_j, χ_k with $1 \leq j, k \leq r$, define the inner product

$$\left\langle \chi_j, \chi_k \right\rangle = \frac{1}{|G|} \sum_{g \in G} \overline{\chi_j(g)} \chi_k(g).$$

Then, $\left\langle \chi_j, \chi_k \right\rangle = \delta_{jk}$.

- **Second Orthogonality Relation:** For a fixed $g, h \in G$,

$$\sum_{j=1}^{r} \chi_j(g)\overline{\chi_j(h)} = \begin{cases} 0 & g \text{ and } h \text{ are not conjugate} \\ |Z_G(h)| & g \text{ and } h \text{ are in the same conjugacy class.} \end{cases}$$

Proof of First Orthogonality Relation

To prove the first orthogonality relation, we will be required to go into the realm of dual representations.

Definition: Let (ρ, V) be a representation of G over F. The *dual representation* is the pair (ρ^*, V^*), with ρ^* defined by

$$\rho^*(g)f = f \circ \rho\left(g^{-1}\right).$$

It can be shown that (ρ^*, V^*) is a representation, and that we have $[\rho^*(g)]_{\beta^*} = \left[\rho\left(g^{-1}\right)\right]_{\beta}^{T}$ whenever β is a basis for V and β^* is the corresponding dual basis.[IV] In particular, $\rho^*(g) = \left(\rho\left(g^{-1}\right)\right)^*$, where $T^*\varphi = \varphi \circ T$ is the dual map.

We start by seeing that if $F = \mathbb{C}$ and (ρ, V) is unitarizable, then $\chi_{\rho^*}(g) = \overline{\chi_\rho(g)}$. This follows from the definition of a unitary map, as $A^*A = AA^* = I$ implies that $\left(A^{-1}\right)^T = \overline{A}$. Therefore, we have

$$\begin{aligned} \chi_{\rho^*}(g) &= \operatorname{Tr}(\rho^*(g)) \\ &= \operatorname{Tr}\left(\left[\rho\left(g^{-1}\right)\right]_{\beta}^{T}\right) \\ &= \operatorname{Tr}\left(\overline{[\rho(g)]_{\beta}}\right) \\ &= \overline{\chi_\rho(g)}. \end{aligned}$$

Additionally, recall that if V and W are finite-dimensional F-vector spaces, then there is a natural isomorphism $\varphi\colon V^* \otimes W \to \hom(V, W)$ given by

$$\varphi(f \otimes w)v = f(v)w.$$

Now, suppose (α, V) and (β, W) are representations of the same G over the same field F. Consider now a representation $(\gamma, \hom(V, W))$, defined by

$$\gamma(g)(f) = \beta(g) \circ f \circ \alpha\left(g^{-1}\right)$$

for any $f \in \hom(V, W)$. The map $\varphi\colon V^* \otimes W \to \hom(V, W)$ that implements the isomorphism $V^* \otimes W \cong \hom(V, W)$ induces an isomorphism $(\alpha^* \otimes \beta, V^* \otimes W) \cong (\gamma, \hom(V, W))$.

If G is finite, and $F = \mathbb{C}$, then α is a unitarizable representation, and we may consider the inner product of characters to be given by

$$\begin{aligned} \left\langle \chi_\alpha, \chi_\beta \right\rangle &= \frac{1}{|G|} \sum_{g \in G} \overline{\chi_\alpha(g)}\chi_\beta(g) \\ &= \frac{1}{|G|} \sum_{g \in G} \chi_{\alpha^*}(g)\chi_\beta(g) \\ &= \frac{1}{|G|} \sum_{g \in G} \chi_{\alpha^* \otimes \beta}(g) \\ &= \frac{1}{|G|} \sum_{g \in G} \chi_\gamma(g). \end{aligned}$$

[IV] Recall that if β is a basis, then the dual basis β^* is defined by the family of v_i^* where $v_i^*\left(v_j\right) = \delta_{ij}$.

Lemma: Let (ρ, Z) be a representation of a finite group G over a field F that satisfies $\operatorname{char}(F) \nmid |G|$. Then,

$$\frac{1}{|G|}\sum_{g\in G}\chi_\rho(g) = \dim\left(Z^G\right),$$

where Z^G denotes the set of fixed points of the representation.

Proof. Extend the group homomorphism $\rho\colon G \to \mathrm{GL}(Z)$ to a homomorphism of F-algebras also denoted $\rho\colon F[G] \to \operatorname{end}(Z)$, given by

$$\rho\left(\sum_{g\in G}\lambda_g g\right) = \sum_{g\in G}\lambda_g \rho(g).$$

Consider now the element in $F[G]$ given by

$$\Delta = \frac{1}{|G|}\sum_{g\in G} g.$$

Let $P = \rho(\Delta) \in \operatorname{end}(Z)$. Then, we have

$$\begin{aligned}\frac{1}{|G|}\sum_{g\in G}\chi_\rho(g) &= \frac{1}{|G|}\sum_{g\in G}\operatorname{Tr}(\rho(g))\\ &= \operatorname{Tr}(\rho(\Delta))\\ &= \operatorname{Tr}(P).\end{aligned}$$

Additionally, observe that $h\Delta = \Delta$, meaning $\Delta^2 = \Delta$, or that $P^2 = P$. Thus, we have that the minimal polynomial $\mu_P(x)$ divides $x(x-1)$. In particular, since $\mu_P(x)$ has only linear factors, it follows that P has eigenvalues either equal to 0 or 1, and P is diagonalizable, meaning we may write

$$\begin{aligned}Z &= E_0 \oplus E_1\\ &= \ker(P) \oplus E_1,\end{aligned}$$

so $\operatorname{Tr}(P) = \dim(E_1)$. It suffices to show that $E_1 = Z^G$. For this, we have that $E_1 \subseteq Z^G$ as we have already established that $g\Delta = \Delta$, so it suffices to show that $E_1 \subseteq Z^G$. This follows from the fact that if $z \in Z^G$, then

$$\begin{aligned}Pz &= \frac{1}{|G|}\sum_{g\in G}\rho(g)z\\ &= z.\end{aligned}$$

□

This finally allows us to prove the First Orthogonality Relation.

Proof of First Orthogonality Relation. From the lemma and the characterization of the inner product as described before the lemma, we have

$$\begin{aligned}\left\langle \chi_\alpha, \chi_\beta\right\rangle &= \frac{1}{|G|}\sum_{g\in G}\chi_\gamma(g)\\ &= \dim\left(\hom(V,W)^G\right)\end{aligned}$$

which by extending the vector space $\hom(V,W)$ to a $\mathbb{C}[G]$-module, gives

$$= \dim\left(\hom_{\mathbb{C}[G]}(V,W)\right)$$

and by Schur's Lemma, we get

$$= \begin{cases} 0 & \alpha \ncong \beta \\ 1 & \alpha \cong \beta. \end{cases}$$

□

The Number of Irreducible Representations

Theorem: Let G be a finite group, $K(G)$ the conjugacy classes of G, and $\operatorname{irr}(G)$ the set of equivalence classes of irreducible complex representations of G. Then,

$$|\operatorname{irr}(G)| = |K(G)|.$$

To prove this, we will prove that these quantities are equal to a third quantity.

Theorem: Let G be a finite group. Then,

$$\dim(Z(\mathbb{C}[G])) = |K(G)|.$$

Proof. Let

$$u = \sum_{x \in G} \lambda_x x,$$

where each $\lambda_x \in \mathbb{C}$. Then, we have $u \in Z(\mathbb{C}[G])$ if and only if $ug = gu$ for all $g \in G$, or that $g^{-1}ug = u$ for all $g \in G$. Writing this out, we get

$$\begin{aligned} g^{-1}ug &= g^{-1}\left(\sum_{x \in G} \lambda_x x\right) g \\ &= \sum_{x \in G} \lambda_x \left(g^{-1}xg\right) \\ &= \sum_{x \in G} \lambda_{gxg^{-1}} x \end{aligned}$$

In particular, this means that for all $x \in G$ and all $g \in G$, we have $\lambda_x = \lambda_{gxg^{-1}}$. This means that $\lambda_x = \lambda_y$ whenever x and y are conjugate to each other.

Letting $r = |K(G)|$ and $K_1, \dots, K_r$ be the conjugacy classes of G, we let

$$u_i = \sum_{g \in K_i} g.$$

It follows that $u \in Z(\mathbb{C}[G])$ if and only if $u \in \operatorname{span}(u_1, \dots, u_r)$. Since conjugacy classes are disjoint, it follows that the u_i are linearly independent, so $\dim(Z(\mathbb{C}[G])) = |K(G)|$. □

Now, we will focus on proving the following theorem, which will be quite a bit more involved.

Theorem: We have

$$\dim(Z(\mathbb{C}[G])) = |\operatorname{irr}(G)|.$$

Recall that if R is a ring, and M is a left R-module, then $\operatorname{end}_R(M)$ denotes the ring of endomorphisms of M over R.

Lemma: If R is a left R-module over itself, then $\operatorname{end}_R(R) \cong R^{\mathrm{op}}$, where R^{op} is the opposite ring (i.e., the ring with multiplication given by $a * b = ba$).

The proof is given by using the fact that $\varphi(a) = a\varphi(1)$, and translating that into an isomorphism between $\operatorname{end}_R(R)$ and R^{op} given by mapping $\varphi \mapsto \varphi(1)$.

Lemma: Let M be an R-module, and let $D = \operatorname{end}_R(M)$. Then,

$$\operatorname{end}_R(M^n) \cong \operatorname{Mat}_n(D).$$

Proof. Given any $A = \left(a_{ij}\right)_{ij} \in \operatorname{Mat}_n(D)$, define a map $\varphi_A \colon M^n \to M^n$ by taking

$$\varphi_A \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} = \begin{pmatrix} \sum_{j=1}^n a_{1j}\left(v_j\right) \\ \vdots \\ \sum_{j=1}^n a_{nj}\left(v_j\right) \end{pmatrix}.$$

It can be seen that φ_A is in fact an endomorphism of M^n and the map $A \mapsto \varphi_A$ is injective.

Now, we will show surjectivity. We let $\alpha_i \colon M \to M^n$ be the embedding into coordinate i, and let $\pi_j \colon M^n \to M$ be the projection onto coordinate j. Observe that we have

$$\begin{aligned} \operatorname{id}_{M^n} &= \sum_{i=1}^n \alpha_i \circ \pi_i \\ \begin{pmatrix} m_1 \\ \vdots \\ m_n \end{pmatrix} &= \sum_{i=1}^n \alpha_i(m_i). \end{aligned}$$

If $\varphi \in \operatorname{end}_R(M^n)$, then we have

$$\begin{aligned} \varphi\begin{pmatrix} m_1 \\ \vdots \\ m_n \end{pmatrix} &= \sum_{i=1}^n \alpha_i \circ \pi_i \circ \varphi\left(\sum_{j=1}^n \alpha_j\left(m_j\right)\right) \\ &= \sum_{i,j=1}^n \alpha_i \circ \pi_i \circ \varphi \circ \alpha_j\left(m_j\right). \end{aligned}$$

Defining $a_{ij} = \pi_i \circ \varphi \circ \alpha_j \in \operatorname{end}_R(M)$, then we have

$$\begin{aligned} \varphi\begin{pmatrix} m_1 \\ \vdots \\ m_n \end{pmatrix} &= \sum_{i,j=1}^n \alpha_i \circ a_{ij}\left(m_j\right) \\ &= \sum_{i=1}^n \alpha_i\left(\sum_{j=1}^n a_{ij}\left(m_j\right)\right) \\ &= \begin{pmatrix} \sum_{j=1}^n a_{1j}\left(m_j\right) \\ \vdots \\ \sum_{j=1}^n a_{nj}\left(m_j\right) \end{pmatrix} \\ &= \left(a_{ij}\right)_{i,j}\begin{pmatrix} m_1 \\ \vdots \\ m_n \end{pmatrix}. \end{aligned}$$

Therefore, $\varphi = \varphi_A$. □

We will apply these two lemmas to prove that $\dim(Z(\mathbb{C}[G])) = |\operatorname{irr}(G)|$, by applying them both to the ring $R = \mathbb{C}[G]$. Observe that $R^{\mathrm{op}} = \mathbb{C}[G^{\mathrm{op}}] \cong \mathbb{C}[G]$, where the latter isomorphism follows from the fact that inversion is an anti-automorphism of the group. In particular, this means that $R \cong R^{\mathrm{op}} \cong \operatorname{end}_R(R)$.

Proof. By Maschke's Theorem, $R = \mathbb{C}[G]$ is an irreducible module over itself, so we may write

$$R = \bigoplus_{j=1}^{n} V_j,$$

where each V_j is an irreducible submodule. Let W be an irreducible complex representation over G, which we may represent as an module over R.

We claim that there must exist some j such that $V_j \cong W$. For this, fix any element $w_0 \in W$, and consider the map $\psi\colon R \to W$ taking $\psi(r) = rw_0$. This is a nonzero homomorphism of R-modules, meaning there is some j such that $\psi|_{V_j} \neq 0$. In particular, this means we have that $\psi|_{V_j}\colon V_j \to W$ is a nonzero R-module homomorphism, but since both of these modules are irreducible, Schur's Lemma gives that this is an R-module isomorphism.

We now seek to understand $\operatorname{end}_R(R)$. Consider the maps $\alpha_j\colon V_j \to R$ the inclusions and $\pi_j\colon R \to V_j$ the projections. If $\varphi \in \operatorname{end}_R(R)$, then for any $1 \leq j,k \leq n$, we have that $\pi_k \circ \varphi \circ \alpha_j\colon V_j \to V_k$ is an R-module homomorphism, meaning that by Schur's Lemma, this composition is zero if $V_j \ncong V_k$ as V_j and V_k are irreducible.

In particular, we have for each j that

$$\varphi\left(V_j\right) \subseteq \bigoplus\left\{V_k \;\middle|\; 1 \leq k \leq n, V_k \cong V_j\right\}.$$

Recall now that $|\operatorname{irr}(G)|$ denotes the equivalence classes of irreducible complex representations of G. Let $r = |\operatorname{irr}(G)|$, and let $W_1, \dots, W_r$ be representatives of equivalence classes. For each $1 \leq i \leq r$, let

$$\widetilde{W_i} = \bigoplus\left\{V_j \;\middle|\; V_j \cong W_i\right\}.$$

Then, we have

(i) $R \cong \displaystyle\bigoplus_{i=1}^{r} \widetilde{W_i}$;

(ii) each $\widetilde{W_i} \neq 0$ as we know there is at least one j such that $W_i \cong V_j$;

(iii) for any endomorphism of R, we have $\varphi\left(\widetilde{W_i}\right) \subseteq \widetilde{W_i}$.

In particular, (iii) implies that

$$\operatorname{end}_R(R) \cong \prod_{i=1}^{r} \operatorname{end}_R\left(\widetilde{W_i}\right).$$

Therefore, we have

$$\begin{aligned} Z(R) &\cong Z(\operatorname{end}_R(R)) \\ &\cong \prod_{i=1}^{r} Z\left(\operatorname{end}_R\left(\widetilde{W_i}\right)\right). \end{aligned}$$

Observe now that $\widetilde{W_i} \cong W_i^{n_i}$ for some $n_i \in \mathbb{N}$, so that

$$\begin{aligned} \operatorname{end}_R\left(\widetilde{W_i}\right) &\cong \operatorname{end}_R\left(W_i^{n_i}\right) \\ &\cong \operatorname{Mat}_{n_i}(\operatorname{end}_R(W_i)). \end{aligned}$$

Since W_i is an irreducible submodule, Schur's Lemma gives

$$\operatorname{end}_R(W_i) = \{\lambda I \mid \lambda \in \mathbb{C}\}$$

$$\cong \mathbb{C},$$

so that $\operatorname{end}_R\left(\widetilde{W_i}\right) \cong \operatorname{Mat}_{n_i}(\mathbb{C})$. Since

$$\begin{aligned} Z\left(\operatorname{Mat}_{n_i}(\mathbb{C})\right) &= \left\{\lambda I_{n_i} \mid \lambda \in \mathbb{C}\right\} \\ &\cong \mathbb{C}, \end{aligned}$$

we have that

$$\begin{aligned} Z(R) &\cong \prod_{i=1}^{r} Z\left(\operatorname{end}_R\left(\widetilde{W_i}\right)\right) \\ &\cong \mathbb{C}^r, \end{aligned}$$

so $\dim(Z(R)) = r$. □

Field Theory and Galois Theory

The final part of the course is going to be focused on field theory and Galois theory.

Overview of Field Extensions

Definition: A *field extension* is a pair of fields (K, F) with $F \subseteq K$. We denote field extensions as K/F. If we have a sequence of field extensions $F_1 \subseteq F_2 \subseteq \cdots \subseteq F_n$, we write this tower of field extensions as $F_n/F_{n-1}/\cdots/F_1$.

Observe that if K/F is a field extension, then K is a vector space over F.

Definition: The *degree* of a field extension K/F, denoted $[K : F]$, is given by $\dim_F(K)$.

Usually we will be interested in finite-degree field extensions, but in principle, $[K : F]$ is the cardinality of the dimension.

Proposition: If $L/K/F$ is a tower of fields, then L/F is finite if and only if L/K and K/F are finite, with $[L : F] = [L : K][K : F]$.

Proof. We show a stronger statement; namely, if $\{\alpha_i\}_{i\in I}$ is a basis for K/F and $\left\{\beta_j\right\}_{j\in J}$ is a basis for L/K, then $S = \left\{\alpha_i\beta_j\right\}_{i\in I, j\in J}$ is a basis for L/F, with the $\alpha_i\beta_j$ distinct.

For this, we need to show that S spans L/F and that its elements are distinct and linearly independent. We will verify the latter.

Suppose it were false. Then, we have elements $\lambda_{ij} \in F$ such that

(a) finitely many are nonzero;

(b) at least one is nonzero;

(c) $\sum_{i,j} \lambda_{ij}\alpha_i\beta_j = 0$.

This is a double sum, meaning that the latter is equal to

$$\sum_j \left(\sum_i \lambda_{ij}\alpha_i\right)\beta_j = 0.$$

The coefficients are all in K, so they are all equal to 0, meaning that for each j,

$$\sum_i \lambda_{ij}\alpha_i = 0,$$

meaning that for each i, $\lambda_{ij} = 0$, so this contradicts the aforementioned assertions. □

If K/F is a field extension, and $S \subseteq K$ is some subset, then $F(S) \subseteq K$ is the smallest subfield of K containing both F and S. If S is finite with elements $\alpha_1, \dots, \alpha_n$, then we write $F(S) = F(\alpha_1, \dots, \alpha_n)$.

Explicitly,we have

$$F(S) = \bigcup_{\substack{T \subseteq S \\ T \text{ finite}}} F(T),$$

and if $S = \{\alpha_1, \dots, \alpha_n\}$ is finite, then

$$F(S) = \left\{ \frac{p(\alpha_1, \dots, \alpha_n)}{q(\alpha_1, \dots, \alpha_n)} \middle| p, q \in F[x_1, \dots, x_n], q(\alpha_1, \dots, \alpha_n) \neq 0 \right\}.$$

We will focus on simple extensions, which are ones of the form $F(\alpha)/F$.

If K/F is a field extension, and $\alpha \in K$, we let

$$V(\alpha) = \{p \in F[x] \mid p(\alpha) = 0\}.$$

Definition: If $V(\alpha) = \{0\}$, then we say α is *transcendental* over F.

If $V(\alpha) \neq \{0\}$, then α is called *algebraic* over F.

If α is transcendental over F, we observe that $F(\alpha) \cong F(x)$, where $F(x) = \operatorname{frac}(F[x])$ is the field of rational functions over F. The isomorphism emerges from evaluation at α.

If α is algebraic over F, then since $F[x]$ is a principal ideal domain, there is a unique monic polynomial such that

$$V(\alpha) = (\mu_{\alpha,F}(x)).$$

We call $\mu_{\alpha,F}(x)$ the *minimal polynomial* of α over F.

Example:

(i) We have that e is transcendental over $\mathbb{Q}$.

(ii) We have that $\sqrt{2}$ is algebraic over $\mathbb{Q}$, with minimal polynomial given by

$$\mu_{\sqrt{2},\mathbb{Q}}(x) = x^2 - 2.$$

(iii) We have that $\omega = e^{2\pi i/3}$ is algebraic over $\mathbb{Q}$ with minimal polynomial given by

$$\mu_{\omega,\mathbb{Q}}(x) = x^2 + x + 1.$$

Observe that in all cases, the minimal polynomial is always irreducible (almost by definition). Furthermore, any monic irreducible polynomial that vanishes at α is the minimal polynomial.

Theorem: Let K/F be a field extension, $\alpha \in K$ algebraic over F. Then,

(i) $F(\alpha) = F[\alpha]$;

(ii) $F(\alpha) \cong F[x]/(\mu_{\alpha,F}(x))$;

(iii) if $n = \deg_F(\alpha)$, then the elements $\left\{1, \alpha, \dots, \alpha^{n-1}\right\}$ form a basis for $F(\alpha)/F$.

Proof. Observe that $F[\alpha] \cong F[\alpha]/(\mu_{\alpha,F}(x))$ by the first isomorphism theorem on the evaluation homomorphism. Since $F[x]$ is a PID and $\mu_{\alpha,F}(x)$ is a field, it follows that $F[\alpha]$ is a field, meaning that $F(\alpha) \subseteq F[\alpha] \subseteq F(\alpha)$, so we have equality.

This means that the map $F[x]/(\mu_{\alpha,F}(x)) \to F[\alpha]$ given by $\overline{\mu(x)} \mapsto \mu(\alpha)$ is an isomorphism, so since $\left\{1, \overline{x}, \dots, \overline{x}^{n-1}\right\}$ forms a basis for $F[x]/(\mu_{\alpha,F}(x))$, we have that $\left\{1, \alpha, \dots, \alpha^{n-1}\right\}$ forms a basis for $F[\alpha] = F(\alpha)$. $\square$

Remark: If K/F is a field extension, and $\alpha \in K$, then the following are equivalent:

(i) α is algebraic over F;

(ii) $F(\alpha)/F$ is finite;

(iii) there exists a subfield $F \subseteq L \subseteq K$ such that $\alpha \in L$ and L/F is finite.

Algebraic Extensions and Algebraic Closures

Definition: An extension K/F is called algebraic if, for any $\alpha \in K$, we have α is algebraic over F.

If α is algebraic over F, then we define

$$\begin{aligned}\deg_F(\alpha) &= \deg(\mu_{\alpha,F}(x))\\ &= [F(\alpha):F].\end{aligned}$$

We start by discussing *relative* algebraic closure before we prove the existence of algebraic closure for arbitrary fields.

Definition: Let K/F be a field extension. The set

$$\operatorname{cl}_K(F) = \{\alpha \in K \mid \alpha \text{ is algebraic over } F\}$$

is known as the relative algebraic closure of F in K.

For instance, $\operatorname{cl}_{\mathbb{C}}(\mathbb{Q}) = \mathbb{A}$, the algebraic numbers.

Theorem: The set $\operatorname{cl}_K(F) \subseteq K$ is a subfield.

Proof. Let $\alpha, \beta \in \operatorname{cl}_K(F)$ with $\beta \neq 0$. We will show that $\alpha \pm \beta$, $\alpha\beta$, and α/β are in $\operatorname{cl}_K(F)$.

Let $L = F(\alpha, \beta)$. Then, all four of these elements are in L. It remains to show that L/F is a finite extension. We observe that $F(\alpha,\beta) = F(\alpha)(\beta)$, giving the tower of extensions $F \subseteq F(\alpha) \subseteq F(\alpha,\beta)$, so that

$$\begin{aligned}[L:F] &= [F(\alpha,\beta):F(\alpha)][F(\alpha):F]\\ &< \infty,\end{aligned}$$

as α is algebraic over F and β, being algebraic over F, is algebraic over $F(\alpha)$. □

Now, we want to investigate the relation between these algebraic extensions and the minimal polynomials $\mu_{\alpha,F}$, $\mu_{\beta,F}$, etc.

A remark on the proof is that the computation gives that

$$\begin{aligned}[F(\alpha,\beta):F] &= \deg_F(\alpha)\deg_{F(\alpha)}(\beta)\\ &\leq \deg_F(\alpha)\deg_F(\beta).\end{aligned}$$

We can elaborate on this.

Corollary: Let K/F be a field extension, and let $\alpha_1, \dots, \alpha_n \in K$ be algebraic over F. Then,

(i) $[F(\alpha_1,\dots,\alpha_n):F] \leq \prod_{i=1}^{n}[F(\alpha_i):F]$;

(ii) $\deg_F(\alpha_i) \mid [F(\alpha_1,\dots,\alpha_n):F]$;

(iii) if $\deg_F(\alpha_i)$ are coprime, then the inequality in (i) is actually an equality.

Example: The splitting field $K \subseteq \mathbb{C}$ of the polynomial $p(x) = x^3 - 2$ is given by adjoining $\mathbb{Q}$ with all the roots of p — i.e., $K = \mathbb{Q}\left(\sqrt[3]{2}, \omega\sqrt[3]{2}, \omega^2\sqrt[3]{2}\right)$, where ω is a primitive cube root of unity.

Yet, we may also write

$$K = \mathbb{Q}\left(\sqrt[3]{2}, \omega\right),$$

as the extension $Q\left(\sqrt[3]{2}, \omega\right)$ is contained in the previous expression for K, while $\omega = \frac{\omega\sqrt[3]{2}}{\sqrt[3]{2}}$.

In particular, $\deg_{\mathbb{Q}}\left(\sqrt[3]{2}\right) = 3$ and $\deg_{\mathbb{Q}}(\omega) = 2$, which are coprime, meaning that $[K : \mathbb{Q}] = 6$.

Lemma: Let $L/K/F$ be a tower of fields where L/K and K/F are algebraic. Then, L/F is algebraic.

Proof. Let $\alpha \in L$. Since α is algebraic over K, there is $p(x) \in K[x]$, given by

$$p(x) = \sum_{i=0}^{n} c_i x^i$$

such that $p(\alpha) = 0$.

Then, we have $p(x) \in F(c_0, \dots, c_n)[x]$, meaning α is algebraic over $F(c_0, \dots, c_n)$. Since K/F is algebraic, we have $[F(c_0, \dots, c_n) : F] < \infty$, with

$$[F(\alpha, c_0, \dots, c_n) : F(c_0, \dots, c_n)] < \infty,$$

meaning that α is algebraic over F. □

Definition: A field K is called algebraically closed if any nonzero polynomial over K has a root in K.

Lemma: Let K be a field. The following are equivalent:

(i) K is *not* algebraically closed;

(ii) there exists a nonconstant $p(x) \in K[x]$ that does not split into a product of linear factors;

(iii) there exists an irreducible polynomial $p(x) \in K[x]$ with $\deg(p) \geq 2$;

(iv) there exists a nontrivial finite extension L/K;

(v) there exists a nontrivial algebraic extension L/K.

Proof. To see that (iv) implies (v), note that finite extensions are algebraic.

For (iv) implies (iii), if $\alpha \in L$ is algebraic with $\alpha \notin K$, then $\mu_{\alpha,K}(x)$ is irreducible with $\deg_K(\alpha) \geq 2$, as else we would have $\alpha \in K$.

For (iii) implies (iv), observe that $K[x]/(p(x))$ is a field that is a finite dimensional vector space over K, so it forms a finite extension.

The equivalence between (i), (ii), and (iii) follows from the fact that irreducible polynomials in $K[x]$ do not split into linear factors (and vice versa), and that a field being not algebraically closed implies the existence of a polynomial that does not have a root in K (hence does not factor into linear terms). □

Definition: Let F be a field. An algebraic closure of F is an algebraically closed field K such that $F \subseteq K$ and the extension K/F is algebraic.

Theorem: Every field F has an algebraic closure.

This is a classic Zorn's Lemma argument, but it requires a bit of care. Ideally, we would consider the set of all pairs of fields (K, F) with K/F algebraic, and partially order this set by inclusion. Then, we would be able to use Zorn's Lemma to prove that there exists maximal algebraic extension M/F and show that M is algebraically closed.

Yet, this has one problem: the notion of "set of all fields" is not logically sound. That is, fields are a proper class rather than a set. The solution will be to fix a set S of "sufficiently large" cardinality with $F \subseteq S$, and consider pairs (K, F) with $K \subseteq S$, ordering $K_1 \leq K_2$ if and only if $K_1 \subseteq K_2$ and $K_2|_{K_1} = K_1$.

An important set-theoretic fact is that an infinite set A has $|A \times A| = |A|$. Furthermore, if a set K is finite, then any algebraic closure L/K is countable, as there are countably many elements of $K[x]$, whence $|L| \leq \max(\aleph_0, |K|)$.

Theorem: Every field has an algebraic closure.

Proof. We will write fields as (K, λ), where K is a set and λ is a field structure on K. Let (F, λ_0) be a field for which we want to construct an algebraic closure.

Fix a set S such that $F \subseteq S$ and $|S| > \max(|F|, \aleph_0)$. Applying Zorn's Lemma on the set

$$\Omega = \{(K, \lambda) \mid F \subseteq K \subseteq S,\ \lambda \text{ is a field structure on } K,\ \lambda|_F = \lambda_0, \text{ and } (K, \lambda)/(F, \lambda_0) \text{ is a field extension}\}$$

partially ordered by the relation $(K, \lambda) \leq (K', \lambda')$ whenever $K \subseteq K'$ and $\lambda'|_K = \lambda$, we find our desired maximal element (M, λ).

Next, we will show that (M, λ) is an algebraic closure of (F, λ_0). By construction, (M, λ) is an algebraic extension, so we only need to show that it is algebraically closed.

Suppose (M, λ) were not algebraically closed, so there exists an algebraic extension (M', λ') with $M \subsetneq M'$. Since M' is a finite extension of M, M' is algebraic over F, and

$$\begin{aligned} \left|M' \setminus M\right| &\leq \left|M'\right| \\ &\leq \max(|F|, \aleph_0) \\ &< |S|, \end{aligned}$$

so since $|M| < |S|$, we have $|S \setminus M| = |S|$, whence $|M' \setminus M| \leq |S \setminus M|$ gives an injection that extends to $\iota\colon M' \to S$. Thus, (M', λ') has an isomorphic copy in S, which violates the maximality of (M, λ). □

Next, we concern ourselves with the uniqueness of algebraic closures.

Definition: Let K/F and L/F be field extensions. A map $\varphi\colon K \to L$ is called an F-embedding (resp. F-isomorphism) if

(i) φ is a ring homomorphism (resp. isomorphism);

(ii) $\varphi|_F = F$.

Recall that there are no nonzero proper ideals in a field, meaning that any nonzero ring homomorphism of fields is automatically injective.

If $\sigma\colon K \to L$ is any field embedding (not necessarily an embedding with respect to a common subfield), then there is an extension to a ring homomorphism $\sigma_*\colon K[x] \to L[x]$ by

$$\sigma_*\left(\sum_{i=0}^{n} a_i x^i\right) = \sum_{i=0}^{n} \sigma(a_i) x^i.$$

Lemma (Simple Extension Lemma)**:** Suppose we have a field embedding $\sigma\colon M \to L$, extensions M'/M and L'/L, and algebraic elements $\alpha \in M'$ and $\beta \in L'$. The following are equivalent:

(i) there exists a field embedding $\sigma_1\colon M(\alpha) \to L(\beta)$ with $\sigma_1|_M = \sigma$ and $\sigma_1(\alpha) = \beta$;

(ii) $\sigma_*(\mu_{\alpha,M}(x)) = \mu_{\beta,L}(x)$.

Proof. For (i) implies (ii), we observe that if there exists such a field embedding, then this induces an injective homomorphism $M[x]/(\mu_{\alpha,M}(x)) \to L[x]/\left(\mu_{\beta,L}(x)\right)$ by pushing forward σ_1. Since $\sigma_1(\alpha) = \beta$, this corresponds to $(\sigma_1)_*(\mu_{\alpha,M}(x)) = \mu_{\beta,L}(x)$.

In the reverse direction since α is algebraic over x, it follows that any element of $M(\alpha)$ is given by $p(\alpha)$ for some $p \in M[x]$. Define

$$\sigma_1(p(\alpha)) = (\sigma_* p)(\beta).$$

This is well-defined since if $p(\alpha) = q(\alpha)$, then $(p - q)(\alpha) = 0$, so $\mu_{\alpha,M}(x) | (p - q)(x)$. Since σ_* is a ring homomorphism, we have $\sigma_*(\mu_{\alpha,M}) | \sigma_*(p) - \sigma_*(q)$. Therefore, we have $(\sigma_* p)(\beta) = (\sigma_* q)(\beta)$. □

Lemma (Embedding Lemma)**:** Let K/F and L/F be extensions such that K/F is algebraic and L is algebraically closed. Then, there is an F-embedding $\sigma\colon K \to L$.

Proof. Apply Zorn's Lemma to the set

$$\{(M,\varphi) \mid M \text{ is an intermediate field extension } F \subseteq M \subseteq K, \text{ with } \varphi\colon M \to L \text{ an } F\text{-embedding}\}$$

partially ordered by $(M,\varphi) \preceq (M',\varphi')$ if and only if $M \subseteq M'$ and $\varphi'|_M = \varphi$. Note that if the maximal element M is not equal to K, then there is some algebraic element $\alpha \in K \setminus M$ and the simple extension lemma can be used to contradict maximality. □

Theorem (Uniqueness of Algebraic Closure)**:** If K and L are algebraic closures of F, then there is an F-isomorphism $K \to L$.

Proof. By the Embedding Lemma, there is an F-embedding $\sigma\colon K \to L$. All we need to show now is that $\sigma(K) = L$.

Since L/F is algebraic and $\sigma(K) \supseteq F$, we have $L/\sigma(K)$ is algebraic.

Since $\sigma(K) \cong K$, and K is algebraically closed, we must have $L = \sigma(K)$ since algebraically closed fields have no nontrivial algebraic extensions. □

We usually denote the algebraic closure of F by $\overline{F}$.

Note that if K/F is an algebraic extension, then $\overline{K} = \overline{F}$.

Splitting Fields, Normal and Separable Extensions

Definition: Let K/F be a field extension and let $\Omega \subseteq F[x] \setminus F$ be a collection of nonconstant polynomials. We say K is a *splitting field* for Ω if

(i) every polynomial $f(x) \in \Omega$ splits over K — that is,

$$f(x) = a\prod_{i=1}^{n}(x - \alpha_i);$$

(ii) we have

$$K = F(\{K\text{-roots of polynomials in } \Omega\}).$$

Example: If $F = \mathbb{Q}$ and $\Omega = \left\{x^3 - 2\right\}$, then over $\overline{\mathbb{Q}}$, we have

$$x^3 - 2 = \left(x - \sqrt[3]{2}\right)\left(x - \zeta_3\sqrt[3]{2}\right)\left(x - \zeta_3^2\sqrt[3]{2}\right),$$

where $\zeta_3 = e^{2\pi i/3}$ is a primitive cube root of unity. Then, the field

$$K = \mathbb{Q}\left(\sqrt[3]{2}, \zeta_3\right)$$

is a splitting field for Ω over $\mathbb{Q}$.

Proposition (Existence and Uniqueness of Splitting Fields)**:** Let F be a field, and let Ω be a family of nonconstant polynomials over F.

(i) For any algebraic closure $\overline{F}$ of F, there exists a unique splitting field K for Ω over F with $K \subseteq \overline{F}$.

(ii) Any splitting field of Ω is contained in some algebraic closure of F.

(iii) A splitting field for Ω over F is unique up to an F-isomorphism.

Proof.

(i) Take $K = F(A)$, where A is the set of all $\overline{F}$-roots of polynomials in Ω. This is the unique such K.

(ii) This is the special case of the observation that any algebraic extension is contained in an algebraic closure.

(iii) We apply (i) and (ii) along with the uniqueness of algebraic closures up to F-isomorphism.

□

Next, we move towards the central concepts in Galois theory.

Definition: Let K/F be an algebraic extension, and let $\overline{F}$ be such that $K \subseteq \overline{F}$. The extension K/F is called *normal* if any of the following equivalent conditions hold

(i) for all irreducible polynomials $p(x) \in F[x]$, if $p(x)$ has a root in K, then $p(x)$ splits in K;

(ii) there exists $\Omega \subseteq F[x] \setminus F$ such that K is the unique splitting field for Ω over F in $\overline{F}$;

(iii) for any F-embedding $\sigma\colon K \to \overline{F}$, we have $\sigma(K) = K$.

Example:

(i) The field $\mathbb{Q}\left(\sqrt[3]{2}, \zeta_3\right)$ is a normal extension of $\mathbb{Q}$ as it is the splitting field for $x^3 - 2$ over $\mathbb{Q}$

(ii) The field $\mathbb{Q}\left(\sqrt[3]{2}\right)$ is not a normal extension of $\mathbb{Q}$, since $p(x) = x^3 - 2$ has a root in $\mathbb{Q}\left(\sqrt[3]{2}\right)$ but does not split into linear factors.

Theorem: The conditions (i) through (iii) in the definition of a normal extension are in fact equivalent.

Proof. Let K/F be an extension such that (i) holds. Define Ω by

$$\Omega = \{\mu_{\alpha,F} \mid \alpha \in K\}.$$

Then, if

$$A = \left\{\overline{F}\text{-roots of polynomials in } \Omega\right\},$$

we have $K = F(A)$.

Suppose (iii) holds. Let $p(x) \in F[x]$ be irreducible, and let α, β be $\overline{F}$-roots with $\alpha \in K$. We want to show that $\beta \in K$.

Without loss of generality, we may assume that $p(x)$ is monic and irreducible, so we have $\mu_{\alpha,F} = \mu_{\beta,F}$. If $\sigma\colon F \to F$ is the identity map, then $\sigma_*\colon F[x] \to F[x]$ is the identity map, so by the simple extension lemma, we have an F-embedding $\tau\colon F(\alpha) \to F(\beta) \subseteq \overline{F}$.

Arguing as in the proof of the main embedding lemma, we may extend to $\tau_1 : K \to \overline{F}$ by taking $\tau_1(\alpha) = \beta$. Applying (iii) to $\sigma = \tau_1$ has $\tau(K) = K$, so since $\tau_1(\alpha) = \beta$, it follows that $\beta \in K$.

Finally, suppose (ii) holds. Any F-embedding $\sigma\colon K \to \overline{F}$ must then permute elements of A, so $\sigma(A) = A$, whence

$$\begin{aligned}\sigma(K) &= \sigma(F(A))\\ &= F(\sigma(A))\\ &= F(A)\\ &= K.\end{aligned}$$

□

The other segment in the definition of a Galois extension is separability.

Definition: Let F be a field, and let $f(x) \in F[x]$ be nonconstant. We say $f(x)$ is separable if f does not have repeated linear factors in $\overline{F}$. That is,

$$f(x) = a\prod_{i=1}^{n}(x - \alpha_i)$$

with α_i distinct.

Proposition: A polynomial $f(x) \in F[x]$ is separable if and only if $\gcd(f(x), f'(x)) = 1$, where $f'(x)$ denotes the formal derivative of f.

Proof. Suppose f is not separable. Write

$$f(x) = (x-\alpha)^2 g(x)$$

for some $\alpha \in \overline{F}$ and $g(x) \in \overline{F}[x]$. Then,

$$f'(x) = (x-\alpha)(2g(x) + (x-\alpha)g'(x)).$$

Therefore, since $(x-\alpha)$ divides both f and f', it follows that $\gcd(f, f') \neq 1$. □

Suppose now that $f(x)$ is irreducible in $F[x]$. Then, since $\deg(f'(x)) < \deg(f(x))$, it follows automatically that $f(x)$ is not separable if and only if $f'(x) = 0$. This is not an issue in characteristic zero, but it is an issue in characteristic p. If $f'(x) = 0$, then f has nonzero coefficients exclusively in degrees that divide p, meaning that $f(x) = g(x^p)$ for some $g(x) \in F[x]$.

Definition: Let K/F be a field extension.

(i) We say an algebraic $\alpha \in K$ is separable if $\mu_{\alpha,F}(x) \in F[x]$ is separable.

(ii) If K/F is algebraic, then K is called separable if $\alpha \in K$ is separable over F for any arbitrary α.

Notice that all algebraic extensions in characteristic zero are separable.

There is a way to characterize separability in terms of F-embeddings, similar to how we characterized normality in terms of F-embeddings.

Definition: Let K/F be an algebraic extension with $K \subseteq \overline{F}$. For any $\alpha \in K$, define the separability degree $\operatorname{sdeg}_F(\alpha)$ to be the number of distinct $\overline{F}$-roots of $\mu_{\alpha,F}$.

Denote by $\operatorname{emb}_F\left(K, \overline{F}\right)$ the set of all F-embeddings from K into $\overline{F}$.

Notice that if $K = F(\alpha)$, then $\sigma \in \operatorname{emb}_F\left(K, \overline{F}\right)$ is determined entirely by $\sigma(\alpha)$, and maps to any of the other roots of $\mu_{\alpha,F}(x)$. In particular, we have

$$\begin{aligned}\left|\operatorname{emb}_F\left(K, \overline{F}\right)\right| &= \operatorname{sdeg}_F(\alpha)\\ &\leq [K:F].\end{aligned}$$

There is a comparable extension of this result to arbitrary finite extensions.

Proposition: Let K/F be a finite extension, and let $\overline{F} \subseteq K$. Then,

(a) $\left|\operatorname{emb}_F\left(K, \overline{F}\right)\right| \leq [K:F]$;

(b) the following are equivalent:

(i) K/F is separable;

(ii) $K = F(\alpha_1, \dots, \alpha_n)$, where each $\alpha_i \in K$ is separable over F;

(iii) there exists a chain of subfields

$$F = K_0 \subseteq K_1 \subseteq \cdots \subseteq K_n = K$$

and elements $\alpha_i \in K_i$ with $1 \leq i \leq n$ such that $K_i = K_{i-1}(\alpha)$ and each α_i is separable over K_{i-1};

(iv) $\left|\operatorname{emb}_F\left(K, \overline{F}\right)\right| = [K:F]$.

Proof. We start by showing that (i) implies (ii) and (ii) implies (iii) in (b).

First, by definition, every element of K is separable, so if $\alpha_1, \dots, \alpha_n$ are a basis for K/F, then $K = F(\alpha_1, \dots, \alpha_n)$ with each α_i separable. This shows (i) implies (ii).

Similarly, if we define $K_i = F(\alpha_1, \dots, \alpha_i)$, then $K_i = K_{i-1}(\alpha)$, and since $\mu_{\alpha,K_{i-1}} | \mu_{\alpha,F}$, it follows that $\mu_{\alpha,K_{i-1}}$ is separable, whence α_i is separable over K_{i-1}. This shows (ii) implies (iii).

Now, let K_i and α_i be defined as in (iii), but we need not assume separability of the α_i. An F-embedding of $K = K_n$ into $\overline{F}$ can be constructed as follows.

Start with the inclusion $\sigma_0 \colon F \to \overline{F}$. Extend via the simple extension lemma to $\sigma_1 \colon K_1 \to \overline{F}$, $\sigma_2 \colon K_2 \to \overline{F}$, etc. up to $\sigma_n \colon K_n \to \overline{F}$.

Counting degrees, we see that if $\sigma_{i-1} \colon K_{i-1} \to \overline{F}$ is chosen, then an extension of σ_{i-1} to σ_i is uniquely determined by its value at α_i, and the valid choices are the roots of $(\sigma_{i-1})_*\left(\mu_{\alpha_i,K_{i-1}}\right)$. These roots are given by the separable degree of α_i over K_{i-1}, whence

$$\left|\operatorname{emb}_F\left(K, \overline{F}\right)\right| = \prod_{i=1}^{n} \operatorname{sdeg}_{K_{i-1}}(\alpha_i),$$

which combined with the fact that $\operatorname{sdeg}_{K_{i-1}}(\alpha_i) \leq [K_i : K_{i-1}]$, gives (a), and yields (iv) in the case of separable extensions.

Finally, for (iv) implies (i), we have that $\operatorname{sdeg}_{K_{i-1}}(\alpha_i) = \deg_{K_{i-1}}(\alpha_i)$ for any choices of α_i and K_i. Since α_1 is always separable over F, and α_1 is arbitrary over F, this gives our desired result. □

Another major fact about finite separable extensions is that they are always generated by a single element. This is known as the Primitive Element Theorem.

Theorem (Primitive Element Theorem)**:** If K/F is a finite separable extension, then there exists $\gamma \in K$ such that $K = F(\gamma)$.

Proof. First, we suppose F is finite, so K is also finite. Then, $K^\times$ is cyclic, so there exists γ such that $K^\times = \langle \gamma \rangle$, implying that $F(\gamma) \supseteq \{0\} \cup \langle \gamma \rangle = K$.

Now, if F is infinite, then $K = F(\alpha_1, \dots, \alpha_n)$ for some $\alpha_1, \dots, \alpha_n$. It therefore suffices to prove the theorem for the case of $n = 2$, as induction provides the rest.

Suppose $K = F(\alpha, \beta)$. Let $m = [K : F]$. Since K/F is separable, there exist exactly m F-embeddings of K into $\overline{F}$, which we call $\sigma_1, \dots, \sigma_m$.

We claim that there exists some $c \in F$ such that the elements $\{\sigma_i(\alpha + c\beta) \mid 1 \leq i \leq m\}$ are all distinct. Since F is infinite, it suffices to show that the set of all $c \in F$ such that $\sigma_i(\alpha + c\beta) = \sigma_j(\alpha + c\beta)$ is finite. In this scenario, we have $\sigma_i(\alpha) + c\sigma_i(\beta) = \sigma_j(\alpha) + c\sigma_j(\beta)$. If it were the case that $\sigma_i(\beta) = \sigma_j(\beta)$, then we would have that $\sigma_i(\alpha) = \sigma_j(\beta)$, implying that $\sigma_i = \sigma_j$ as maps since they are defined solely by where they map α and β. If $\sigma_i(\beta) \neq \sigma_j(\beta)$, then we may solve for c by taking

$$c = \frac{\sigma_j(\alpha) - \sigma_i(\alpha)}{\sigma_i(\beta) - \sigma_j(\beta)},$$

which has only finitely many options.

We obtain the primitive element theorem by selecting $\gamma = \alpha + c\beta$ as in the claim. Then, for all $1 \leq i \leq m$, we have $\sigma_i|_{F(\gamma)}$ is an F-embedding of $F(\gamma)$ into $\overline{F}$, with $\sigma_i(\gamma) \neq \sigma_j(\gamma)$ for all $i \neq j$. Therefore, we have

$$\begin{aligned}
[F(\gamma) : F] &\geq \left|\operatorname{emb}_F\left(F(\gamma), \overline{F}\right)\right| \\
&\geq m \\
&= [K : F] \\
&\geq [F(\gamma) : F].
\end{aligned}$$

□

Galois Extensions and Galois Groups

Definition: Given a field extension K/F, we define its automorphism group to be

$$\begin{aligned}\operatorname{aut}(K/F) &= \{F\text{-automorphisms of } K\}\\ &= \{\sigma\colon K \to K \mid \sigma \text{ is a ring isomorphism, } \sigma|_F = \mathrm{id}_F\}.\end{aligned}$$

If an extension K/F is algebraic with algebraic closure $\overline{F}$ containing K, we recall that $\operatorname{emb}_F\left(K,\overline{F}\right)$ is the set of all F-embeddings of K into $\overline{F}$.

We may then identify $\operatorname{aut}(K/F)$ as the set

$$\operatorname{aut}(K/F) = \left\{\sigma\colon \operatorname{emb}_F\left(K,\overline{F}\right) \,\middle|\, \sigma(K) = K\right\}.$$

We have

$$\begin{aligned}|\operatorname{aut}(K/F)| &\leq \left|\operatorname{emb}_F\left(K,\overline{F}\right)\right|\\ &\leq [K:F],\end{aligned}$$

with equality if and only if K/F is Galois, as

$$\operatorname{aut}(K/F) = \operatorname{emb}_F\left(K,\overline{F}\right)$$

if and only if K/F is normal and

$$\left|\operatorname{emb}_F\left(K,\overline{F}\right)\right| = [K:F]$$

if and only if K/F is separable.

Definition: An algebraic extension K/F is called Galois if it is normal and separable.

Observe that this means a finite extension K/F is Galois if and only if $|\operatorname{aut}(K/F)| = [K:F]$.

For Galois extensions K/F, we will refer to $\operatorname{aut}(K/F)$ as $\operatorname{Gal}(K/F)$, and call it the *Galois group* of the extension.

Observe that if K/F is an algebraic extension, then K/F is Galois if and only if K is a splitting field over F of some family of separable polynomials over F. Note that if $\operatorname{char}(F) = 0$ or F is a finite field, then the separability assumption can be removed.

Definition: Let F be a field with algebraic closure $\overline{F}$, and let $\alpha \in \overline{F}$. The roots of $\mu_{\alpha,F}(x)$ are called the Galois conjugates of α over F, denoted $\operatorname{conj}(\alpha)$.

Example:

(a) If $F = \mathbb{Q}$ and $\alpha = 2+i$, then $\mu_{\alpha,F}(x) = (x-2)^2+1$, meaning that

$$\operatorname{conj}(\alpha) = \{2-i, 2+i\}.$$

(b) If $F = \mathbb{Q}$ and $\alpha = 1+\sqrt{2}$, then $\mu_{\alpha,F}(x) = (x-1)^2-2$, whence

$$\operatorname{conj}(\alpha) = \left\{1-\sqrt{2}, 1+\sqrt{2}\right\}$$

Now, if K/F is a Galois extension, then for any $\alpha \in K$ and any $\sigma \in \operatorname{Gal}(K/F)$, we have $\sigma(\alpha) \in \operatorname{conj}(\alpha)$. Additionally, for any $\beta \in \operatorname{conj}(\alpha)$, $\operatorname{conj}(\beta) = \operatorname{conj}(\alpha)$.

It follows that $\sigma(\operatorname{conj}(\alpha)) = \operatorname{conj}(\alpha)$, meaning that the Galois group acts on the Galois conjugates by permutations. There is thus a homomorphism $\varphi_\alpha\colon \operatorname{Gal}(K/F) \to \operatorname{sym}(\operatorname{conj}(\alpha))$.

Example:

(i) If $F = \mathbb{Q}$ and $K = \operatorname{spl}_{\mathbb{Q}}\left(x^3 - 2\right)$ is the splitting field, then we have that $\alpha = \sqrt[3]{2}$ has three Galois conjugates, meaning there is an injective homomorphism $\varphi_\alpha\colon \operatorname{Gal}(K/F) \to S_3$. Yet, since $[K : F] = |\operatorname{Gal}(K/F)| = 6$, it follows that this is actually a surjective homomorphism, so that $\operatorname{Gal}(K/F) \cong S_3$.

The generators are given by $\sqrt[3]{2}$, $\zeta_3\sqrt[3]{2}$, and $\zeta_3^2\sqrt[3]{2}$, where ζ_3 denotes a cube root of unity.

(ii) If $F = \mathbb{Q}$ and $K = \operatorname{spl}_{\mathbb{Q}}\left(x^4 - 2\right)$, then for $\alpha = \sqrt[4]{2}$, we have that $K = \mathbb{Q}\left(\sqrt[4]{2}, i\right)$ and an injective homomorphism $\operatorname{Gal}(K/F) \to S_4$. This is not a surjective homomorphism since $[K : F] = |\operatorname{Gal}(K/F)| = 8$. However, we can still uniquely describe this Galois group, as the only subgroups of order 8 in S_4 are isomorphic copies of D_4, following from the Sylow theorems as $24 = 2^3 \cdot 3$ and all Sylow 2-subgroups are conjugate.

Now, we turn our attention to perhaps the defining result in Galois theory: the correspondence between intermediate subfields and subgroups.

By a(n) (intermediate) subfield of K/F, we mean a subfield $F \subseteq L \subseteq K$. Whenever H is a subgroup of the Galois group $G = \operatorname{Gal}(K/F)$, we denote by K^H the *fixed field* of H, defined by

$$K^H := \{\alpha \in K \mid \sigma(\alpha) = \alpha \text{ for all } \sigma \in H\}.$$

Theorem (Galois Correspondence)**:** Let K/F be a finite Galois extension with $G = \operatorname{Gal}(K/F)$. Then, the maps

$$\begin{aligned} \Phi\colon \{\text{subfields of } K/F\} &\to \{\text{subgroups of } G\} \\ L &\mapsto \operatorname{Gal}(K/L) \\ \Psi\colon \{\text{subgroups of } G\} &\to \{\text{subfields of } K/F\} \\ H &\mapsto K^H \end{aligned}$$

are mutually inverse and define an inclusion-reversing correspondence.

We will prove this theorem by using Artin's Lemma.

Theorem (Artin's Lemma)**:** Let K be a field, and let G be a finite subgroup of $\operatorname{aut}(K)$, with $F = K^G$. Then,

(i) K/F is Galois;

(ii) K/F is finite;

(iii) $\operatorname{Gal}(K/F) = G$ as a subgroup of $\operatorname{aut}(K)$.

First, we show how Artin's Lemma implies the Galois correspondence.

Proof of Galois Correspondence assuming Artin's Lemma. Let $H \leq G$ be a subgroup. Then,

$$\begin{aligned} \Phi \circ \Psi(H) &= \Phi\left(K^H\right) \\ &= \operatorname{Gal}\left(K/K^H\right) \\ &= H \end{aligned}$$

by Artin's Lemma.

Now, let $F \subseteq L \subseteq K$, and set $M = \Psi \circ \Phi(L)$. Then, we have

$$M = K^{\operatorname{Gal}(K/L)}.$$

Since L is contained in the fixed field of $\operatorname{Gal}(K/L)$, it follows that $L \subseteq M$. To show that $L = M$, it suffices to show that $[M : L] = 1$, which since

$$[M : L] = \frac{[K : L]}{[K : M]},$$

is equivalent to showing that $[K : L] = [K : M]$. Applying Artin's Lemma to $\operatorname{Gal}(K/L)$, we have $\operatorname{Gal}(K/M) = \operatorname{Gal}(K/L)$, whence $[K : L] = [K : M]$, so $L = M$. $\square$

Now, we can prove Artin's Lemma.

Proof of Artin's Lemma.

(i) We will show that for any $\alpha \in K$, the minimal polynomial $\mu_{\alpha,F}$ is separable and splits completely over K. Let $\alpha \in K$, and let O_α be the orbit of α under G. Write

$$O_\alpha = \{\alpha_1, \dots, \alpha_m\},$$

where $m = |O_\alpha|$, and set

$$p(x) = \prod_{i=1}^{m} (x - \alpha_i).$$

We claim that $p(x) \in F[x]$. Observe that if $\sigma \in \operatorname{aut}(K)$, and $\sigma_* \in \operatorname{aut}(K[x])$ is defined by

$$\sigma_*\left(\sum_{i=0}^{n} b_i x^i\right) = \sum_{i=0}^{n} \sigma(b_i) x^i,$$

then the claim amounts to showing that $\sigma_* p(x) = p(x)$ for all $\sigma \in G$. This immediately follows from the fact that the α_i are defined by the orbit of α, as

$$\begin{aligned}\sigma_* p(x) &= \prod_{i=1}^{m} (x - \sigma(\alpha_i)) \\ &= p(x).\end{aligned}$$

Therefore, we must have $\mu_{\alpha,F}(x) | p(x)$, but since the roots of $p(x)$ are distinct, it follows that $\mu_{\alpha,F}(x)$ is separable and splits completely over K.

(ii) and (iii) We will show that $[K : F] \leq |G|$. This will show that K/F is a finite extension, and since $G \subseteq \operatorname{Gal}(K/F)$, we have

$$\begin{aligned}[K : F] &\leq |G| \\ &\leq |\operatorname{Gal}(K/F)| \\ &\leq [K : F],\end{aligned}$$

giving equalities everywhere. First, let $\alpha \in K$, and observe that

$$\begin{aligned}[F(\alpha) : F] &= \deg(\mu_{\alpha,F}(x)) \\ &\leq \deg(p(x)) \\ &= |O_\alpha| \\ &\leq |G|.\end{aligned}$$

Select the maximal such α. If $F(\alpha) = K$, then we are done. Else, let $\beta \in K \setminus F(\alpha)$. Then, $F(\alpha, \beta)/F$ is a finite separable extension since K/F is already separable.

By the Primitive Element Theorem, it follows that there is $\gamma \in K$ such that $F(\alpha, \beta) = F(\gamma)$, which contradicts maximality as $F(\alpha) \subsetneq F(\gamma)$, so that $[F(\gamma) : F] > [F(\alpha) : F]$.

□

Finite Fields

Given a prime p, we let $\mathbb{F}_p$ denote the field $\mathbb{Z}/p\mathbb{Z}$, and $\overline{\mathbb{F}_p}$ the algebraic closure of $\mathbb{F}_p$.

Theorem (Existence and Uniqueness of Finite Fields)**:** Let p be a prime, and let

$$\mathbb{F}_{p^n} := \left\{ x \in \overline{\mathbb{F}_p} \;\middle|\; x^{p^n} - x = 0 \right\} >$$

The following hold:

(i) $\mathbb{F}_{p^n}$ is a field of order p^n and it is the unique subfield of order p^n in $\overline{\mathbb{F}_p}$;

(ii) if F is any finite field of characteristic p, then $|F| = p^n$ for some n, and $F \cong \mathbb{F}_{p^n}$.

Proof.

(i) To verify that it is a field, observe that it is closed under addition

$$\begin{aligned}(x+y)^{p^n} &= \sum_{k=0}^{p^n} \binom{p^n}{k} x^k y^{p^n-k} \\ &= x^{p^n} + y^{p^n} \\ &= x + y,\end{aligned}$$

where the second line follows from the fact that we are in characteristic p, and the definitions

$$\begin{aligned}(xy)^{p^n} &= x^{p^n} y^{p^n} \\ &= xy \\ \left(x^{-1}\right)^{p^n} &= \left(x^{p^n}\right)^{-1} \\ &= x^{-1}\end{aligned}$$

yield the desired result.

Next, taking derivatives, we observe that if $p(x) = x^{p^n} - x$, then $p'(x) = -1$, so that $\gcd(p, p') = 1$, whence $p(x)$ is separable, so that $\left|\mathbb{F}_{p^n}\right| = p^n$.

For uniqueness, suppose F is any subfield of $\overline{\mathbb{F}_p}$ with $|F| = p^n$. We will show that $F \subseteq \mathbb{F}_{p^n}$. The group $\mathbb{F}^\times$ has order $p^n - 1$, meaning that for all $x \in F^\times$, we have

$$x^{p^n-1} - 1 = 0,$$

whence $x^{p^n} - x = 0$, so $F \subseteq \mathbb{F}_{p^n}$.

(ii) Since we are in characteristic p, it follows that F contains a copy of $\mathbb{F}_p$ generated by 1. Furthermore, since F is finite, we have $\left[F : \mathbb{F}_p\right] = p^{n-1}$ for some n, so that $|F| = p^n$. There thus exists an embedding $F \to \overline{\mathbb{F}_p}$, whence $F \cong \mathbb{F}_{p^n}$.

□

Proposition: Let $m, n \in \mathbb{N}$. Then, $\mathbb{F}_{p^m} \subseteq \mathbb{F}_{p^n}$ if and only if $m|n$.

Proof. If $\mathbb{F}_{p^m} \subseteq \mathbb{F}_{p^n}$, then $\mathbb{F}_{p^n}$ is a vector space over $\mathbb{F}_{p^m}$ with $d = \left[\mathbb{F}_{p^n} : \mathbb{F}_{p^m}\right]$ satisfying

$$\left|\mathbb{F}_{p^n}\right| = p^{md}.$$

In the reverse direction, the base case $d = 1$ is vacuous, so suppose we have $x^{p^{md}} = x$. Then, $x^{p^{m(d+1)}} = x^{p^m} = x$, whence $x \in \mathbb{F}_{p^{m(d+1)}}$. □

Remark: We can observe that

$$\overline{\mathbb{F}_p} = \bigcup_{n \geq 1} \mathbb{F}_{p^n}.$$

Corollary: If F is a finite field of order p^n, then F has unique subfields of order p^m for each $m|n$.

Definition: Let K be any field of characteristic p. The map $\mathrm{fr}_K : K \to K$ given by $x \mapsto x^p$ is known as the Frobenius map.

Observe that if the characteristic is equal to p, then fr is an endomorphism, and it is injective since it is a nonzero endomorphism of a field; therefore, if K is finite, then fr is an automorphism.

Theorem: Let K/F be a finite extension of finite fields. Then,

(i) K/F is Galois;

(ii) $\mathrm{Gal}(K/F)$ is cyclic;

(iii) if $|F| = p^m$ with p prime, then $\mathrm{Gal}(K/F) = \langle \mathrm{fr}_K^m \rangle$.

Proof. Without loss of generality, we may assume $K \subseteq \overline{\mathbb{F}_p}$. Then, we have $K = \mathbb{F}_{p^n}$, and $m|n$ with $F = \mathbb{F}_{p^m}$. Let

$$\mathrm{fr} := \mathrm{fr}_{\overline{\mathbb{F}_p}}.$$

We observe that

$$\begin{aligned} \mathbb{F}_{p^r} &= \left\{ x \in \overline{\mathbb{F}_p} \;\middle|\; x^{p^s} = x \right\} \\ &= \mathrm{fix}(\mathrm{fr}^s). \end{aligned}$$

In particular, this means

$$(\mathrm{fr}_K)^m = (\mathrm{fr}^m)|_K$$

is an automorphism of K that fixes $F = \mathbb{F}_{p^m}$.

Therefore, $\mathrm{fr}_K^m \in \mathrm{Gal}(K/F)$. The order of fr_K^m is the smallest d such that $\mathrm{fr}_K^{md} = \mathrm{id}_K$. Therefore, we have

$$\begin{aligned} K &\subseteq \mathrm{fix}\left(\mathrm{fr}_K^{md}\right) \\ &= \mathbb{F}_{p^{md}}. \end{aligned}$$

The smallest such d is given by $d = \frac{n}{m}$, so that $\frac{n}{m} = [K : F]$. Therefore, we have $\langle \mathrm{fr}_K^m \rangle \leq \mathrm{Gal}(K/F)$, but since

$$\begin{aligned} [K : F] &= \left| \langle \mathrm{fr}_K^m \rangle \right| \\ &\leq |\mathrm{Gal}(K/F)| \\ &\leq [K : F]. \end{aligned}$$

Therefore, these are equalities, and we have $\mathrm{Gal}(K/F) = \langle \mathrm{fr}_K^m \rangle$. □

Remark: The group $\mathrm{Gal}\left(\overline{\mathbb{F}_p}/\mathbb{F}_p\right)$ contains $\left\langle \mathrm{fr}_{\overline{\mathbb{F}_p}} \right\rangle \cong \mathbb{Z}$, but is significantly larger. It is given by the *profinite completion* of $\mathbb{Z}$, denoted $\hat{\mathbb{Z}}$.